

**“IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON
PULMONARY FUNCTIONS, CHEST EXPANSION AND SIX
MINUTE WALK DISTANCE IN INDIVIDUAL WITH CHRONIC
OBSTRUCTIVE PULMONARY DISEASE”**

BY

ANITA MADANLAL PRAJAPAT

DISSERTATION SUBMITTED TO

GUJARAT UNIVERSITY

AHMEDABAD, GUJARAT,

IN PARTIAL FULFILLMENT OF THE REQUIREMENT,

FOR THE DEGREE OF

MASTER OF PHYSIOTHERAPY

IN

CARDIO-RESPIRATORY DISORDER

AHMEDABAD INSTITUTE OF MEDICAL SCIENCES

NEAR KARNAVATI EYE HOSPITAL, LAPKAMAN,

AHMEDABAD-380 060



2023-2025

**“IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON
PULMONARY FUNCTIONS, CHEST EXPANSION AND SIX
MINUTE WALK DISTANCE IN INDIVIDUAL WITH CHRONIC
OBSTRUCTIVE PULMONARY DISEASE”**

BY

ANITA MADANLAL PRAJAPAT

DISSERTATION SUBMITTED TO

GUJARAT UNIVERSITY

AHMEDABAD, GUJARAT,

IN PARTIAL FULFILLMENT OF THE REQUIREMENT,

FOR THE DEGREE OF

MASTER OF PHYSIOTHERAPY

IN

CARDIO-RESPIRATORY DISORDER

UNDER THE GUIDANCE OF

DR. HARDINI PRAJAPATI,

SENIOR LECTURER & PG GUIDE

MPT (CARDIOPULMONARY SCIENCES)

AHMEDABAD INSTITUTE OF MEDICAL SCIENCES NEAR

KARNAVATI EYE HOSPITAL, LAPKAMAN,

AHMEDABAD-380 060,



2023-2025

DECLARATION BY THE CANDIDATE

I hereby declare that this dissertation entitled, **“IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON PULMONARY FUNCTIONS, CHEST EXPANSION AND SIX MINUTE WALK DISTANCE IN INDIVIDUAL WITH CHRONIC OBSTRUCTIVE PULMONARY DISEASE”** is a bonafide and genuine research work carried out by **ANITA MADANLAL PRAJAPAT** under the guidance of **DR. HARDINI PRAJAPATI, MPT (CARDIOPULMONARY SCIENCES) SENIOR LECTURER & PG GUIDE** at **AHMEDABAD INSTITUTE OF MEDICAL SCIENCES**.

DATE:

ANITA MADANLAL PRAJAPAT

PLACE: AHMEDABAD

©COPYRIGHT

AHMEDABAD INSTITUTE OF MEDICAL SCIENCES

DECLARATION BY THE CANDIDATE

I here declare that the Gujarat university, Ahmedabad shall have all rights to preserve, use or disseminate this dissertation entitled “**IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON PULMONARY FUNCTIONS, CHEST EXPANSION AND SIX MINUTE WALK DISTANCE IN INDIVIDUAL WITH CHRONIC OBSTRUCTIVE PULMONARY DISEASE**” is a bonafide and genuine research work carried out by **ANITA MADANLAL PRAJAPAT** under the guidance of **DR. HARDINI PRAJAPATI, MPT (CARDIOPULMONARY SCIENCES) SENIOR LECTURER & PG GUIDE** at **AHMEDABAD INSTITUTE OF MEDICAL SCIENCES**. The **Gujarat university, Ahmedabad** shall all rights to preserve, use or disseminate this dissertation in print or electronic format for academic/research purpose.

DATE:

ANITA MADANLAL PRAJAPAT

PLACE: AHMEDABAD

©GUJARAT UNIVERSITY, AHMEDABAD

ENDORSEMENT BY THE PRINCIPAL AND HEAD OF
INSTITUTE

This is to certify that the dissertation entitled, “**IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON PULMONARY FUNCTIONS, CHEST EXPANSION AND SIX MINUTE WALK DISTANCE IN INDIVIDUAL WITH CHRONIC OBSTRUCTIVE PULMONARY DISEASE**” is a bonafide and genuine research work carried out by **ANITA MADANLAL PRAJAPAT** under the guidance of **DR. HARDINI PRAJAPATI, MPT (CARDIOPULMONARY SCIENCES) SENIOR LECTURER & PG GUIDE** at **AHMEDABAD INSTITUTE OF MEDICAL SCIENCES**

DR. T. KANNA AMARNATH
MPT (ORTHOPAEDIC DISORDER)
PRINCIPAL AND PG GUIDE,
Ahmedabad institute of medical sciences,
Lapkaman, ahmedabad-380060

DATE:

PLACE: AHMEDABAD

CERTIFICATE BY THE GUIDE

This is to certify that the dissertation entitled, “**IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON PULMONARY FUNCTIONS, CHEST EXPANSION AND SIX MINUTE WALK DISTANCE IN INDIVIDUALS WITH CHRONIC OBSTRUCTIVE PULMONARY DISEASE**” submitted in partial fulfilment of the degree of Master of Physiotherapy is a Bonafide work carried out by **ANITA MADANLAL PRAJAPAT**. Under my guidance and supervision. This is to certify that no part of this dissertation has been submitted to any University/Institute. It is her original contribution. I recommend that work be considered for evaluation leading to the award of degree of **Master of Physiotherapy**.

DR. HARDINI PRAJAPATI

MPT (CARDIOPULMONARY SCIENCES)

SENIOR LECTURER & PG GUIDE

Ahmedabad institute of medical sciences,

Lapkaman, ahmedabad-380060

DATE:

PLACE: AHMEDABAD

ACKNOWLEDGEMENT

“શ્રી ગણેશાય નમઃ”

This journey has been a mixture of many ups and downs and yet so overwhelming. As we say the path is more difficult than the destination, so as the journey is Coming to an end, I would like to offer my heartfelt gratitude to all who stood by my side.

Words might fall short to express my gratitude to the people who have been part of this gracious journey of mine and supporting, accompanying and encouraging me at every point of journey.

Before starting any good work, God blessings are always needed. I am pleased to thank The God who laid the foundation of knowledge and wisdom, for showering blessings throughout the journey and showering me the true path of success with hard work.

I am extremely grateful to my dearest parents **MR. MADANLAL PRAJAPAT** and **MRS. SHARDA PRAJAPAT** who sacrificed, remained self-less, made me believe in myself and stood by my side always. Words will always fall a short for their immense love, care and support they have showered on me. I would also like to thank my younger brother **MOHIT PRAJAPAT** for helping me in best possible manner.

I would express my deepest gratitude to my respected guide **DR. HARDINI PRAJAPATI** Ma'am for always supporting and providing invaluable guidance throughout the research. It was a great privilege and honor to work under your guidance. Your dynamic, Vision, serenity and motivation have deeply inspired me.

I would like to thank to my Hon'ble Principal **Dr. KANNA AMARNATH** sir for his invaluable guidance, **Dr. NEETA VYAS Ma'am**, for her immense Support and encouragement, Hon'ble Vice Principal **Dr. ALPA PUROHIT** Ma'am for her continuous guidance and support. And my sincere thanks to all **FACULTY MEMBERS** of AIMS and Institutional Research and Ethics Committee Members for their valuable contribution when needed.

I would thank **Dr. HEMANT TIWARI** Sir, Assistant Professor in statistics, NHL Medical College, Amdavad for providing us statistical information and calculations required in research studies.

Thank you, **Dr. DHAVAL DARJI**, for your unwavering support and for sharing your knowledge, which has greatly enriched this research.

I would extend sincere gratitude towards my cardio colleagues **PRAKRUTI GAMIT** and **KANCHI MEHTA** for their support and guidance regarding the study.

I am exceptionally grateful to my best friends to my DOST, **UMESH SUTHAR AND POOJA SUTHAR** and my husband **AAKASH PRAJAPATI** for being there throughout in all my thick and thins and for being my pillars of strength.

My sincere thanks to most important people without whom my master's journey is incomplete **LEELA CHOUDHARY, VASVI VORA, MAULI SHAH, PARSHVI DOSHI, PRAPTI RAVAL, EKTA SONI, and BHOOMI SOLANKI**

Last but not the least, I would like to thank the pillars of AIMS college ie. whole AIMS family who provided me the best possible hep throughout the journey.

Thanks to all subjects for their participation in the study and other who all have contributed directly or indirectly during my dissertation

DATE:

ANITA MADANLAL PRAJAPAT

PLACE: AHMEDABAD

DEDICATED TO
THE ALMIGHTY GOD,
MY PARENTS & BROTHER
MY BETTER HALF,
MY IN-LAWS,
RESPECTED TEACHERS
&
FRIENDS

**“IMMEDIATE EFFECT OF CHEST WALL KINESIO
TAPING ON PULMONARY FUNCTIONS, CHEST
EXPANSION AND SIX MINUTE WALK DISTANCE IN
INDIVIDUAL WITH CHRONIC OBSTRUCTIVE
PULMONARY”**

**MASTER OF PHYSIOTHERAPY
IN
CARDIO-RESPIRATORY DISORDER**



2023-2025

INDEX

SR NO.	TITLE	PAGE NO.
1.	INTRODUCTION	1
2.	NEED OF THE STUDY	6
3.	AIMS AND OBJECTIVES	7
4.	HYPOTHESIS	8
5.	REVIEW OF LITERATURE	9
6.	REVIEW OF ARTICLES	14
7.	MATERIALS AD METHODOLOGY	30
8.	RESULTS	41
9.	DISCUSSION	53
10.	CONCLUSION	57
11.	CLINICAL IMPLICATION	58
12.	LIMITATIONS AND FUTURE STUDIES	59
13.	SUMMARY	60
14.	REFERENCES	62
15.	ANNEXURES	I

LIST OF TABLES

SR.NO	TITLE	PAGE NO.
1.	DEMOGRAPHIC DETAILS OF THE PARTICIPANTS	41
2.	STATISTICAL TEST FOR OUTCOME MEASURES	42
3.	EFFECT OF KINESIO-TAPING ON FEV ₁	43
4.	EFFECT OF KINESIO-TAPING ON FVC	44
5.	EFFECT OF KINESIO-TAPING ON FEV ₁ /FVC	45
6.	EFFECT OF KINESIO-TAPING ON PEFR	46
7.	EFFECT OF KINESIO-TAPING ON 6MWD	47
8.	EFFECT OF KINESIO-TAPING ON MCE	48
9.	EFFECT OF KINESIO-TAPING ON SPO ₂	49
10.	EFFECT OF KINESIO-TAPING ON RESPIRATORY RATE	50
11.	EFFECT OF KINESIO-TAPING ON HEART RATE	51
12.	EFFECT OF KINESIO-TAPING ON MEAN BLOOD PRESSURE	52

LIST OF GRAPHS

SR.NO	TITLE	PAGE NO.
1.	PRE AND POST EFFECT OF FEV ₁	43
2.	PRE AND POST EFFECT OF FVC	44
3.	PRE AND POST EFFECT OF FEV ₁ /FVC	45
4.	PRE AND POST EFFECT OF PEFR	46
5.	PRE AND POST EFFECT OF 6MWD	47
6.	PRE AND POST EFFECT OF MCE	48
7.	PRE AND POST EFFECT OF SPO ₂	49
8.	PRE AND POST EFFECT OF RR	50
9.	PRE AND POST EFFECT OF HEART RATE	51
10.	PRE AND POST EFFECT OF MEAN BLOOD PRESSURE	52

LIST OF FIGURES

SR NO.	TITLE	PAGE NO.
1.	CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)	1
2.	ANATOMY OF RESPIRATORY TRACK	2
3.	NEURAL CONTROL OG BREATHING	10
4.	GOLD CRITERIA FOR COPD	14
5.	VOLUME TIME AND FLOW TIME CURVE	23
6.	LUNG VOLUME AND CAPACITIES	24
7.	PFT MACHINE	33
8.	KINESIO TAPING & CONSERT FORM	33
8.1	SPHYGMOMANOMETER AND PULSE OXIMETER	33
9.	PERFORMING PULMONARY FUNCTION TEST	35
10.	MEASURING SIX MINUTE WALK DISTANCE	36
11.	MEASURING MIDDLE CHEST EXPANSION	37
12.	MEASURING VITALS	38
13.	PERFORMING KINESIO TAPING	39

ABBREVIATIONS

WHO – World Health Organization

COPD – Chronic Obstructive pulmonary disease

PFT – Pulmonary Function Test

FEV₁ – Forced Expiratory Volume in 1 sec

FVC – Forced Vital Capacity

PEFR – Peak Expiratory Flow Rate

6MWD – Six Minute Walk Distance

MCE – Middle Chest Expansion

RR – Respiratory Rate

HR – Heart Rate

KT – Kinesio Taping

MBP – Mean Blood Pressure

BMI -Body Mass Index

INTRODUCTION

INTRODUCTION

Chronic obstructive pulmonary disease is a progressive condition characterized by airflow limitation which is not fully reversible. (WHO).^[1]

By 2024, COPD is expected to be the third greatest cause of mortality globally, with 90% of deaths occurring in poor and middle-income nations. Worldwide the most frequent risk factor for COPD is smoking, however air pollution from burning wood and biomass fuels has also been linked to the disease in several countries.^[1]

Chronic obstructive pulmonary disease is a leading source of illness and mortality worldwide.^[1] The term COPD typically includes **chronic bronchitis** and **emphysema**. Asthma is also one type of chronic obstructive disease classified separately as it is reversible and mostly triggered by allergens or irritants.^[2]

A. Chronic bronchitis is defined as a cough that produces copious sputum on most days for at least three continuous months each year for at least two years.^[3]

B. Emphysema, also known as alveolar or parenchyma disease, causes permanent swelling of air spaces at the terminal non-respiratory bronchioles and damaging changes to the alveolar wall.^[3]

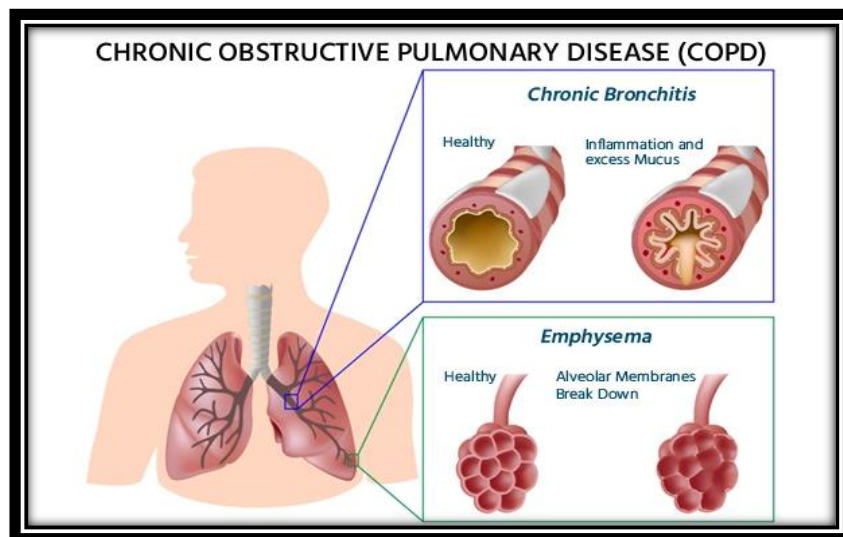


Fig 1: CHRONIC OBSTRUCTIVE PULMONARY DISEASE

These two disorders cause distinct COPD patterns that can be clinically identified. Both patterns present symptoms such as dyspnoea, chronic cough, and expectoration.^[4]

Chronic inflammation leads to structural changes, constriction of the tiny airways, degradation of the lung parenchyma, and loss of alveolar elasticity, reducing lung elastic recoil. Inflammation narrows the airways, causing premature closure of expiration causes air trapping in the distal airways, resulting in hyperinflation of the lungs.^[5]

The impact of treatment on disease development is measured using criteria such as PFT (Pulmonary function test), chest expansion, and exercise tolerance test ^[3]. Dyspnoea Grading, Spirometry is a reliable lung function test that should be performed on all individuals with COPD.^[7]

In patients with COPD, changes in the PFT test are seen Reduced FEV1 (force expiratory volume in 1st second), Reduced FVC (force vital capacity) and Decreased FEV/FVC ratio,

Reduced TLC (Total lung capacity),

Increased RV (residual volume),

Reduced DLCO (Diffusion capacity).^[3, 5, 8]

Decreased PEFR and chest expansion is also seen in COPD patients as they have reduced lung capacities due to hyperinflation. These all measurements showed fluctuations in normal value due to disease progression and also due to effect of the treatment.^[6]

Although COPD cannot be cured, but symptoms can be managed by medication, surgery, and physical therapy.^[7]

Pharmacologic treatment options include bronchodilators (β -agonists, anticholinergics, and corticosteroids), antibiotics, and vaccines. Etc.^[9]

Surgical therapy includes lung volume reduction surgery (LVRS), which removes nonfunctional, overextended lung tissues.

Non-pharmacological management for COPD includes physiotherapy, ventilation assistance, and oxygen therapy.^[10]

The primary goal of rehabilitation is to lessen symptoms, enhance quality of life, and increase physical and emotional engagement in daily activities. Pulmonary rehabilitation consists of exercise training, nutritional counselling and education

Physiotherapy management strategies include relaxation, breathing exercises, airway clearance techniques, positive pressure breathing, respiratory muscle training, pulmonary rehabilitation, and aerobic training.^[10]

These exercises can help enhance mucus clearance, minimize work of breathing, and increase respiratory muscle strength and endurance. [5, 11] There are also more traditional medicine strategies that we might utilize as nonpharmacological management for certain medical disorders.

The method of Kinesio tape was developed by Dr. Kenzo Kase, in between 1973 and 1979 with the intention of providing support for the musculoskeletal structure without immobilization and its associated adverse effects.^[12]

Kase commercialized the tape for KT in 1982, with elastic, cohesive, lightweight, and ventilation properties. The initial objective was to manage oedema, provide soft tissue support, preserve joints, and relieve the heat caused by active inflammation.^[12]

However, due to a lack of scientific research, it is still unclear how kinesiology taping may alter respiratory function and provide relief from dyspnoea. The elastic nature of kinesiology tape produces traction, convolutions and mechanical deformation of the skin and superficial layers of viscoelastic connective tissue matrix (fascia) and striated muscle fibres.^[14,15] It is believed to stimulate low threshold mechanoreceptors in the skin, fascia, muscle spindles, and Golgi tendon organs, resulting in an influx of impulses in myelinated nerve fibres, kinaesthetic awareness, muscular tone, and strength all influence proprioceptive perception..^[13,14,15]

Kinesiology taping may also alleviate breathlessness by providing mechanical support to the structures associated with breathing, i.e., the rib cage, upper abdomen and neck where the primary and accessory inspiratory muscles are located.^[13]

**NEED
OF
THE
STUDY**

NEED OF THE STUDY

Cigarette smoking is the major cause of COPD, which causes hyperinflation and increased mucus secretion in the lungs, resulting in dyspnoea and improper gas exchange. These Abnormalities have an impact on the patient's overall health and functioning capacities. Therefore, prevention and treatment are top priorities. Kinesio taping, a form of traditional alternative medicine, has been clinically utilized to alleviate pulmonary problems.

There is a lack of literature on the effectiveness of Kinesio taping in reducing symptoms in COPD patients. This highlights the need for further research into the potential benefits of Kinesio taping in chronic obstructive pulmonary disease. Alternative physiotherapy treatments for pulmonary rehabilitation, include non-pharmacological interventions such as Kinesio taping, forced expiratory technique, deep breathing exercise, etc.

**AIM
AND
OBJECTIVES**

AIM AND OBJECTIVES

AIM:

To investigate the immediate effect of chest wall Kinesio Taping on pulmonary functions and six-minute walk distance in individual with chronic obstructive pulmonary disease.

OBJECTIVES:

• **PRIMARY OBJECTIVES:**

- 1) To study the immediate effect of Kinesio Taping on pulmonary functions by using spirometry with the parameters FEV₁, FVC, FEV₁/FVC, PEF_R in individual with chronic obstructive pulmonary disease.
- 2) To study the immediate effect of Kinesio Taping on six-minute walk distance in individual with chronic obstructive pulmonary disease.

• **SECONDARY OBJECTIVES:**

- 3) To study the immediate effect of Kinesio Taping on middle chest expansion in individual with chronic obstructive pulmonary disease.
- 4) To study the immediate effect of Kinesio Taping on oxygen saturation level (SPO₂) in individual with chronic obstructive pulmonary disease.
- 5) To study the immediate effect of Kinesio Taping on respiratory rate in individual with chronic obstructive pulmonary disease.
- 6) To study the immediate effect of Kinesio Taping on heart rate in individual with chronic obstructive pulmonary disease.
- 7) To study the immediate effect of Kinesio Taping on Mean blood pressure (MBP) in individual with chronic obstructive pulmonary disease.

HYPOTHESIS

HYPOTHESIS

NULL HYPOTHESIS:

[H₀¹]: There is no immediate effect on lung functions (FVC, FEV₁, FEV₁/FVC, and PEFR) with Kinesio taping in individual with chronic obstructive pulmonary disease.

[H₀²]: There is no immediate effect on six-minute walk distance with Kinesio taping in individual with chronic obstructive pulmonary disease.

[H₀³]; There is no immediate effect on chest expansion with Kinesio taping in individual with chronic obstructive pulmonary disease.

ALTERNATE HYPOTHESIS:

[H₁¹]: There is an immediate effect on lung functions (FVC, FEV₁, FEV₁/FVC, and PEFR) with Kinesio taping in individual with chronic obstructive pulmonary disease.

[H₁²]: There is an immediate effect on six-minute walk distance with Kinesio taping in individual with chronic obstructive pulmonary disease.

[H₁³]; There is an immediate effect on chest expansion with Kinesio taping in individual with chronic obstructive pulmonary disease.

**REVIEW
OF
LITERATURE**

REVIEW OF LITERATURE

[A] ANATOMY AND PHYSIOLOGY OF RESPIRATORY SYSTEM:

ANATOMY: -

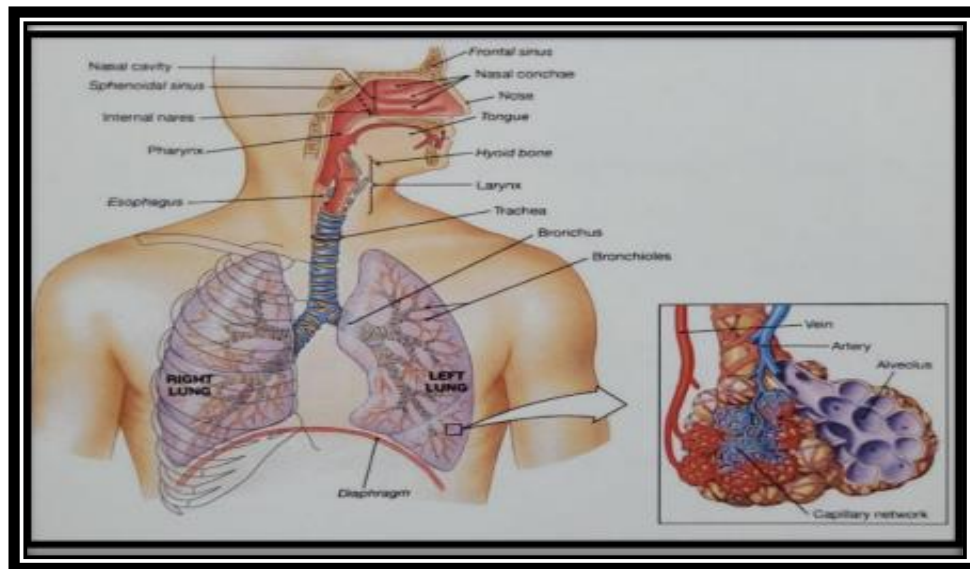


FIGURE -2: ANATOMY OF RESPIRATORY TRACT

Anatomy of respiratory system mainly divided in to two parts **upper respiratory track & lower respiratory track.**^[2]

- **Upper respiratory track involve** - Nose, Nasal cavity, Pharynx, Naso pharynx, Oropharynx, Laryngopharynx, larynx.
- **Lower respiratory track involve-** Trachea, main stem & lobar bronchi, Respiratory bronchioles, alveoli, alveolar duct alveolar sac.

PHYSIOLOGY:

The lungs' primary function is gas exchange. The pulmonary circulation adds oxygen (O_2) and removes excess CO_2 from mixed venous blood.^[16]

The lungs' ability to exchange gases relies on four basic physiological functions.^[17]

- The diaphragm & thoracic muscles must be capable of expanding the thorax and lungs to produce a sub atmospheric pressure.
- The airway must be unobstructed to allow gas to flow in to the lungs and reach the alveoli.
- The cardiovascular system must circulate blood through the lungs and ventilated alveoli. O₂ and CO₂ must be able to diffuse through the alveolar-capillary membrane.

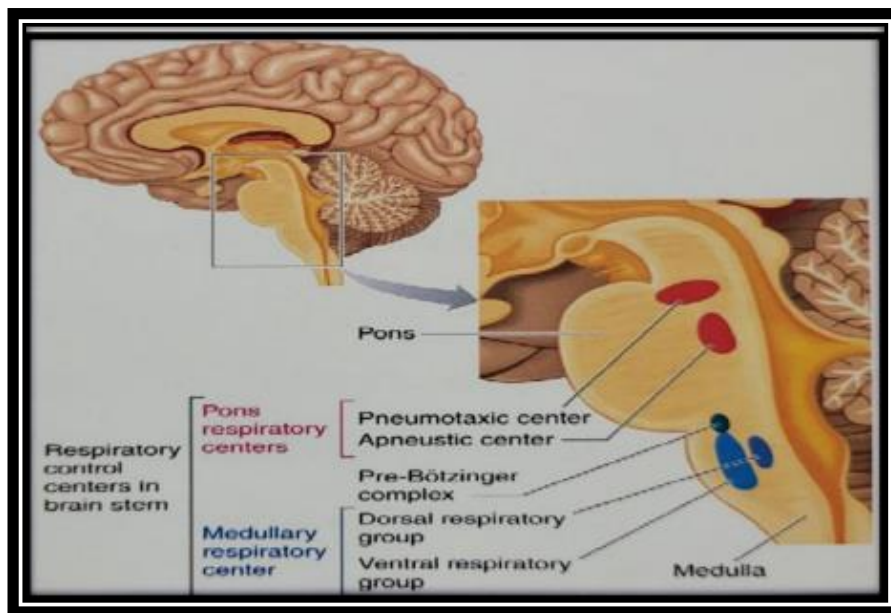


FIGURE -3: NEURAL CONROL OF BREATHING

Control of breathing: - ^[18]

➤ The pattern of respiration is regulated by two mechanisms.

1) Neural mechanism-

- Medullary centres -inspiratory and expiratory control
- Pontine centres - pneumotaxic and apneustic control

2) Chemical mechanism-

- Central chemoreceptors
- Peripheral chemoreceptor

[B] CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

Chronic obstructive pulmonary disease, abbreviated as COPD or occasionally COLD (chronic obstructive lung disease), denotes a pathological condition marked by airflow blockage due to chronic bronchitis or emphysema.^[18] Chronic Obstructive Pulmonary Disease (COPD) is a predominant cause of morbidity and mortality globally, resulting in a significant and escalating economic and social burden.^[8]

According to the new recommendations statement from the **AMERICAN THORACIC SOCIETY about COPD** – ^[18]

“Airflow restriction that is not entirely reversible is a defining feature of Chronic Obstructive Pulmonary Disease (COPD), a disease state that can be prevented and treated. Usually progressive, the airway restriction is linked to an aberrant inflammatory response. Reaction of the lungs to harmful gases or particles, typically brought on by smoking cigarettes. Despite having an impact on the lungs, COPD has serious systemic effects as well.”

Airflow blockage that is not entirely reversible is a hallmark of COPD. Over several months, there is little change in the restriction of airflow, and over time, it typically gets worse. Smoking is the main cause of COPD. COPD may also develop as a result of work exposure. Exacerbations frequently happen when symptoms deteriorate quickly and persistently beyond typical daily fluctuations.^[19]

The following should be used as a definition of COPD-^[19]

- The definition of airflow restriction is a decreased FEV1/FVC ratio such that FEV1/FVC is less than 0.7 (where FEV1 is the forced expired volume in one second and FVC is the forced vital capacity).
- A diagnosis of COPD should only be made if there are respiratory symptoms, such as coughing or dyspnoea, and the FEV1 is more than the 80% expected normal.

The blockage of airflow is caused by injury to the parenchyma and airways. The damage is caused by persistent inflammation, which is typically brought on by tobacco smoke and is distinct from that observed in asthma. Significant blockage of airflow could be existed before the individual knew it.^[19]

Symptoms of COPD include impairment, reduced quality of life, and symptoms that may be addressed by medication and other treatments that have little to no effect on airway blockage. In patients with airflow, the preferred term for the ailment is now COPD, that is blockage in those who had previously been diagnosed with emphysema or chronic bronchitis.^[19]

A diagnosis of COPD is made by clinical judgment based on a mix of history, physical examination, and spirometry to confirm the existence of airflow restriction. There is no single diagnostic test for the condition.^[19]

Pathology & Pathophysiology: -

There are pathological alterations in the lung parenchyma, pulmonary vasculature, proximal and peripheral airways that are typical of COPD.^[20]

Chronic inflammation with a rise in particular inflammatory cell types in various lung regions and structural alterations brought on by repeated injury and healing are examples of the pathological alterations.^[1]

COPD is typified by a particular pattern of inflammation that involves lymphocytes, macrophages, and neutrophils. In addition to interacting with lung parenchyma and airway structural cells, these cells emit inflammatory mediators. Reduce the key findings of FEVI from inflammatory and peripheral airway constriction.^[1]

In order to evaluate chronic airflow blockage, forced expiratory airflow is typically tested. FEV₁ is simply one of several measurements of forced expiratory airflow, and it can be influenced by two main factors: pressure applied and resistance encountered.^[3]

- Lung tissue normally recoils elastically after being released from the stretched position it acquires during inspiration, which results in the lung pressure applied during expiratory airflow.

- The constriction of the airways causes resistance to airflow. Airway thickening, inflammation, increased mucus, or contraction of the bronchial wall (bronchospasm) can all result in narrowing. The most reliable, widely used, and potentially most practical spirometry measurement is FEV₁.^[8]

The gold standard for precise and reliable measurement of lung function is spirometry, according to both national and GOLD international COPD recommendations.^[7]

However, spirometry is only one method of determining the severity of COPD. This technique measures the amount of air that the patient can exhale from their lungs following a maximal inhalation in order to evaluate lung function. The indices that came from this compelled when these results are compared with expected normal values based on age, height, and sex, a measure of the degree of airway blockage can be derived. Exhaled manoeuvre has emerged as the most accurate and trustworthy method of validating a diagnosis of COPD.^[7]

A maximal forced exhale (the biggest effort possible) followed by a maximum deep inhalation (fully full lungs) is the standard spirometry manoeuvre. From this blow, several indices can be obtained.^[7]

- 1) FVC- forced vital capacity (The total volume of air that the patient can forcibly exhale in one breath)
- 2) FEV₁ - forced expiratory volume in 1 second (The volume of air that the patient is able to exhale in the first second of forced expiration)
- 3) FEV₁/FVC - the ratio of FEV₁/FVC expressed as a percentage (Values of FEV₁ & FVC are measured in litres and are also expressed as a percentage of the predicted values for that individual).

Stage	Spirometric Findings
Mild	FEV ₁ /FVC <.70 FEV ₁ ≥ 80% predicted
Moderate	FEV ₁ /FVC <.70 FEV ₁ between 50% and 80% predicted
Severe	FEV ₁ /FVC <.70 FEV ₁ between 30% and 50% predicted
Very Severe	FEV ₁ /FVC <.70 FEV ₁ <30% predicted or FEV ₁ <50% predicted + chronic respiratory failure

Fig 4: GOLD SPIROMETRIC CRITERIA FOR COPD SEVERITY

B1) PREVALENCE: -

B-1.1: -GOLD (2024) Global Initiative for Chronic Obstructive Lung Disease.

The aging of the global population and ongoing exposure to COPD risk factors are expected to contribute to a rise in the prevalence and burden of COPD throughout the ensuing decades. You can get information about the prevalence of COPD on international websites like the World Bank/WHO and the World Health Organization (WHO).

The burden of disease worldwide (GBD). There is growing data that makes it possible to assess the prevalence of COPD more precisely. COPD is significantly more common in smokers and ex-smokers than in non-smokers, in people over 40 than in those under 40, according to a number of systematic reviews and meta-analyses in men as opposed to women. To determine the prevalence and risks of COPD in individuals over 40 worldwide, the Burden of Obstructive Lung Diseases (BOLD) program employed a standardized methodology that included questionnaires and pre- and post-bronchodilator spirometry.^{23 (31-32)} The total prevalence of COPD, according to BOLD, was 11.8% (standard error (SE)).^[1]

B-1.1: - The Minister of State In The Ministry Of Health And Family Welfare Of India 2020.

Chronic Respiratory Diseases, including COPD, were responsible for 10.9% of all deaths and 6.4% of all Disability Adjusted Life, according to the Global Burden of Disease (GBD) research "India: Health of the Nation's States," which was released by ICMR in November 2017 years (DALY). According to a 2018 Lancet journal study, there were an estimated 55.3 million cases of Chronic Obstructive Pulmonary Disease (COPD) in India in 2016. It was estimated that Himachal Pradesh, Uttarakhand, Haryana, and Jammu & Kashmir have high crude prevalences of COPD.^[21]

B-1.2: - Epidemiology COPD in India (2011).

In India, almost 24 million persons over 40 had COPD in 2010. According to data monitors, this number is expected to rise by 34% to about 32 million by 2020. Men are primarily affected by COPD, and only. Over the course of the predicted period, women will account for 40% of all COPD cases in India; demographic shifts will be the main driver of this development.^[22] Chronic obstructive pulmonary disease (COPD) is a disease of the elderly in India, as evidenced by the fact that 8.25 million of the estimated 23.68 million common cases in 2010 occurred in those aged 40 to 60.2. The number of men with COPD in 2010 (about 14.11 million) was 1.5 times that of women (roughly 9.57 million), and this trend is expected to hold true until 2020. The most evident reason for this is the increased in India, men smoke at higher rates than women.^[22]

B2) MORBIDITY & MORTALITY:

B-2.1: -GOLD (2024) Global Initiative for Chronic Obstructive Lung Disease.

Traditionally, hospitalizations, emergency department visits, and doctor visits are used as morbidity measurements. As of right now, research shows that COPD-related morbidity rises with age and that comorbidities start to appear earlier in life in COPD patients. Chronic conditions that are associated with smoking, aging, and/or COPD, such as diabetes mellitus, musculoskeletal

disability, and cardiovascular disease, may also have an impact on morbidity in COPD.

The growing smoking epidemic, decreased mortality from other common causes of death (such as infectious diseases and ischemic heart disease), global population aging, especially in high-income nations, and a lack of effective disease-modifying treatments have been the main causes of COPD-related mortality. Information from the given these limitations, it can be estimated that COPD causes around three million deaths worldwide each year, with the GBD Study 2017 estimating a COPD-attributable mortality rate of 42/100,000 (4.72% of all-cause fatalities). By 2060, it is predicted that over 5.4 million people will die each year from COPD and related diseases due to the rising smoking prevalence in LMICs and the aging of the populations in high-income nations.^[1]

B-2.2: -Wan c. Tan and Tze p. Ng (2008). “COPD in Asia: where East meets West” The epidemic of tobacco exposure and indoor and outdoor air pollution in Asian countries are the main causes of the current higher burden of COPD in Asia compared to developed Western countries, both in terms of the overall number of deaths and the burden of disease as measured in years of life lost and years spent living with disability. One This article's goal was to summarize the existing research on COPD in Asia and to emphasize the ways that COPD in Asia differs from or is comparable to that observed in the West.^[23]

B3) PFT in COPD:

B-3.1: - Nicholas R Anthonisen, John E Connett et al (1994) Effects of Smoking Intervention and the Use of an Inhaled Anticholinergic Bronchodilator on the Rate of Decline of FEV₁ has conducted a study in US and Canada to determine whether a program incorporating smoking intervention and use of an inhaled bronchodilator can slow the rate of decline in forced expiratory volume in 1 second (FEV₁) in smokers aged 35 to 60 years who have mild obstructive pulmonary disease. An aggressive smoking intervention program significantly reduces the age-related decline in FEV, in middle-aged smokers with mild airways obstruction. Use of an 18 inhaled anticholinergic bronchodilator results in a

relatively small improvement in FEV₁, that appears to be reversed after the drug is discontinued. Use of the bronchodilator did not influence the long-term decline of FEV₁.^[24]

B-3.2: - Gavin C. Donaldson et al (2005) did a study about "Airway and Systemic Inflammation and Decline in Lung Function in Patients with COPD." According to his study Patients with COPD experience lower airway and systemic inflammation, and an accelerated decline in FEV₁. There is no evidence on whether this inflammation changes over time, or if it is associated with a faster decline in FEV₁. they found that, Airway and systemic inflammatory markers increase over time in COPD patient; high levels of these markers are associated with a faster decline in lung function.^[25]

B-3.3:-Lee & Fry (2010) "Systematic review of the evidence relating FEV₁ decline to giving up smoking" has examined decline in FEV₁ among never smokers, continued smokers, ex-smokers and quitters (those who discontinued smoking between recruitment and follow up). Never smokers had a lower rate of FEV₁ decline than continued smokers (10.8 ml/year less; 95% confidence interval [CI] 8.9 to 12.8), and the rate of decline of FEV₁ of continued smokers was significantly more than in the other three groups ($p < 0.001$). Among the three non-smoking groups there was no statistically significant difference in FEV₁ decline, suggesting that disease progression in ex-smokers and quitters was more closely aligned with non-smokers than smokers.^[26]

[C] PHYSIOTHERAPY MANAGEMENT & PHYSICAL REHABILITATION IN COPD:

Physiotherapy management of COPD

Main aims of physiotherapy treatment are – ^[27]

- To facilitate the removal of secretion and relieve any bronchospasm.
- To improve the pattern of breathing & breathing control
- To teach local relaxation and improve posture

- To mobilize the thorax & shoulder girdle
- To improve the exercise tolerance

Pulmonary rehabilitation is the primary aspect of physiotherapy management. Reduction of symptoms, enhancement of quality of life, and increased emotional and physical engagement in daily activities are the main objectives of rehabilitation. Exercise training, dietary counselling, and education are all parts of pulmonary rehabilitation.^[1]

Following rehabilitation, COPD patients benefit from increased exercise capacity, a decrease in the perceived severity of dyspnoea, and an improvement in health-related quality of life, per GOLD criteria. lowers the quantity of hospital stays and days in the hospital lessens the anxiety and melancholy brought on by disease, enhances arm function through upper limb strength and endurance training, boost survivability.^[1]

C-1) Garrod R. Lammers et al (2007) Systemic reviews on Role of physiotherapy management of COPD by: - talked about the several interventions that physiotherapists frequently employ, including physical training for the respiratory and peripheral muscles, bronchopulmonary hygiene procedures, and breathing exercises. The evaluations indicate that Inspiratory Muscle Training (IMT) may increase inspiratory muscle strength, and breathing exercises may improve health-related quality of life in asthma. However, the longer-term consequences of breathing exercises on morbidity have not been taken into consideration, nor is the clinical significance of improved respiratory muscle strength per se recognized. According to one review, bronchopulmonary hygiene practices in bronchiectasis and chronic obstructive pulmonary disease (COPD) clearly increase the generation of sputum. Higher sputum production is linked to frequent exacerbations and high bacterial load, indicating that better sputum clearance may have significant therapeutic benefits. It is advised that future research examine the long-term impacts of bronchopulmonary hygiene practices on morbidity.[28]

C-2) Waterhouse et al “Chronic obstructive pulmonary disease. NICE clinical guideline 101” (2010): - Carried out a powered, randomized study with 240 patients (mean age ~69 years) to compare community-based with hospital-based pulmonary rehabilitation, followed by traditional or phone follow-up). They could not find significant difference between the hospital (108.7%) and community (90.95%) rehabilitation groups in terms of the percentage change in the distance walked during an endurance shuttle walk test (mean difference 17.8%; 95% CI - 24.3 to 59.9; $p = 0.405$). The endurance shuttle walk test is intended to walk individuals unassisted at a set speed to ensure they are at 85% of their maximum oxygen capacity [VO₂ max] relative to baseline. Additionally, there was no significant difference between the groups in terms of the lengthening of time they could walk after rehabilitation. The lack of any notable there were still differences in these outcomes at the 6-, 12-, and 18-month post-rehabilitation follow-ups, and there didn't seem to be any evidence of a difference between telephone and traditional follow-up at those times either. The authors also performed an economic analysis, which found no financial benefit to telephone follow-up or any of the rehabilitation settings.^[26]

C-3) Beauchamp et al. (2011) “Optimal duration of pulmonary rehabilitation for individuals with chronic obstructive pulmonary disease - a systematic review” This review looked to figure out how long pulmonary rehabilitation lasted in relation to health-related quality of life and exercise tolerance in people with chronic obstructive pulmonary disease (COPD). The Cochrane Library of clinical trials, Physiotherapy Evidence Database [Pedro], MEDLINE, PubMed, CINAHL, EMBASE, and reference lists of relevant articles were searched in order to find randomized controlled trials (RCTs) comparing various durations of pulmonary rehabilitation in patients with COPD. Two reviewers conducted the searches and used the Pedro and Jadad scales to evaluate the quality of the trials. Five RCTs were deemed eligible. With a range of 3 to 8, the mean Pedro score was 6 and the mean Jadad was 2. Three trials found that the longer programs improved health-related quality of life, while two trials found that longer programs improved exercise ability. Program duration and outcomes varied widely, making it impossible to conduct a meta-analysis of the data. The effects of longer-term pulmonary rehabilitation programs on health-related quality of life in people with

COPD seem to be more favourable; however, the findings on exercise ability are less certain. The paucity of research precludes drawing a more firm judgment about the ideal length of rehabilitation.^[29]

C-4) Grosslink et al (2011) “Impact of inspiratory muscle training in patients with COPD” has the effects of inspiratory muscle training (IMT) in patients with chronic obstructive pulmonary disease (COPD) were examined in a meta-analysis that included 32 randomized controlled studies. In terms of patient characteristics and training type (strength or endurance training, in addition to general exercise training), both overall and subgroup analyses were conducted. There were notable gains in quality of life (+3.8 units), endurance time (+261 s), maximal inspiratory muscle strength (P(I_{max}); +13 cmH₂O), and walking distances of 6 or 12 minutes (+32 and +85 m, respectively). There was a considerable decrease in dyspnoea (Transitional Dyspnoea Index +2.8 units; Borg score -0.9 points). While there were no impacts on maximal exercise capacity, endurance exercise capacity tended to increase. Dyspnoea, functional exercise capacity, and P(I_{max}) were not significantly impacted by respiratory muscle endurance training. When IMT was added to a general exercise program, P(I_{max}) increased significantly, and in patients with inspiratory muscle weakness (P(I_{max}) <60 cmH₂O), functional exercise capacity tended to increase. IMT improves quality of life, dyspnoea, and inspiratory muscle strength and endurance. Inspiratory muscle endurance training was found to be less effective than respiratory muscle strength training. In patients with inspiratory muscle weakness, IMT enhanced P(I_{max}) and tended to improve exercise performance.^[29]

[D] KINESIO TAPING & REASPIRATORY SYSTEM: -

Alternative methods for managing patients with chronic illnesses have been prompted by drug side effects.

Kinesio taping is one of the alternative traditional medicine techniques that we can use as the non-pharmacological management for various respiratory conditions.

Applying elastic, thin, non-medicated, water-resistant adhesive tape to the skin using a range of methods is known as kinesiology taping. In contrast to stiff

tapes, kinesiology tape can be extended longitudinally by at least 60% of its initial length and worn constantly to support joints and soft tissues without limiting movement. [28]

The "**muscle facilitatory**" kinesiology taping technique involves stretching adhesive kinesiology tape and applying it to the skin overlying the muscles of inspiration. It is said that this modifies myofascial tone and muscle contraction by causing tactile stimulation and concentric pull on superficial soft tissues. [13]

Kinesiology tape's elastic properties cause the skin, surface layers of the viscoelastic connective tissue matrix (fascia), and striated muscle fibres to traction, convolutions, and mechanical deformation. [14,15]

This allegedly activates low threshold mechanoreceptors in muscle spindles, fascia, Golgi tendon organs, and skin, causing a barrage of impulses in myelinated Nerve fibres control muscle tone and strength as well as proprioceptive and kinaesthetic perception. [28]

Kinesiology tape can also help with dyspnoea by giving the breathing-related structures the rib cage, upper abdomen, and neck, which house the principal and accessory inspiratory muscles mechanical support. [29]

D 1: - Saniye Aydogan Arslan (2018): - “Effects of Kinesio Taping Applied to Diaphragm Muscle on Aerobic Exercise Capacity and Pulmonary Function in Sedentary Individuals (2018) Turkey”. This study was approved by the University Ethical Committee (Kırıkkale University Clinical Research Ethical Committee; chairperson Mehmet Savas Ekici, MD, Prof.; protocol number 05/05; date of approval 23/2/2016): The purpose of this study was to investigate the short-term effects of diaphragmatic Kinesio taping (KT) on sedentary people's ability to do aerobic activity and pulmonary function. KT applied to the diaphragm muscle improves pulmonary and aerobic function both in the short term and when compared to the sham KT group. [30] Cotton tape is latex-free, thin, and breathable, preventing skin irritation. It is designed to reflect the skin's qualities. The thickness is comparable to the epidermis layer, and the flexibility is close to that of human skin. It can. The stretch up to 55-60% of its length to accommodate for skin stress and relaxation during movement. Tapes can stay on the skin for 3-7 days after application, depending on skin texture and environment. [12]

D 2: Murat Tomruk, PhD (2020) Turkey: -“Effects of thoracic Kinesio taping on pulmonary functions, respiratory muscle strength and functional capacity in patients with chronic obstructive pulmonary disease: A randomized controlled trial”: - Chronic Obstructive Pulmonary Disease (COPD) is characterized by respiratory and peripheral muscle dysfunctions that lead to ventilatory restriction, dyspnoea, and inactivity. These symptoms ultimately reduce functional ability. The rehabilitative procedure known as Kinesio Taping (KT) involves applying a unique stretchy tape, so boosting blood flow and muscle contraction. A total of 27 patients with COPD were involved (16 in the KT group and 11 in the control group). For the KT group, thoracic KT was used to support the respiratory muscles throughout the subcostal region. Both groups were given deep breathing techniques. Two days a week for six weeks, interventions were conducted. The lung spirometer was used to measure the maximal respiratory mouth pressures and function. The Modified Borg Scale and the Modified Medical Research Council Dyspnoea Scale were used to measure the severity of weariness and dyspnoea, respectively. Using the six-minute walk test, functional capacity was assessed. Patients with COPD may benefit from thoracic KT to enhance their functional ability and lung function.^[31]

D 3: John Langendoen 2016: - “Rationales and Evidence of Elastic Taping: A clinician’s perspective”: - Department of Physical Therapy, Dongnam Health University, Jeongja 3- dong, Jangan-gu, Suwon-si, Gyeonggi-do, Republic of Korea This systematic review's goal was to find out how applying tape can improve physical conditions. Among the search tactics used for this review was a literature search conducted by International. Between January 2014 and February 2015, Kinematic Academy used national journals, websites, and CINAHL, Cochrane, Google Scholar, and PubMed in 12 countries. "Kinesio tape, Kinesio tape, Kinesio taping, elastic taping, taping, functional taping, myofascial taping, and sensomotor taping" were among the search terms. All articles, including those published in other languages, were included in the review. There were 821 publications found by these searches. Applying tape has been shown to provide a number of benefits, including increased lymphatic and blood circulation, improved range of motion, increased mechanoreceptor and joint stability, and reduced discomfort. Nobody were complaining about the advantages of taping.

However, it takes more work to find proof of the consequences of applying tape.^[32]

D 4: - Ganesh, B. R.; Praneetha, N.; Naik, Varun (2023) “Comparison of Two Thoracic Kinesio-Taping Techniques on Pulmonary Function, Oxygen Saturation, and Exercise Capacity in COPD Patients: A Randomized Clinical Trial: - Institute of Physiotherapy, Belagavi, Karnataka, India state” that the diaphragmatic and intercostal muscle taping techniques were not very different. It was observed that both methods were equally successful in enhancing exercise capacity, oxygen saturation, and pulmonary function in patients with COPD. Therefore, when used in conjunction with traditional physiotherapy treatment, thoracic Kinesio taping can aid in improving patient results.^[33]

[E] SPIROMETRY-

A pulmonary function test can reveal critical details about each of these crucial separate processes that facilitate gas exchange. Numerous measurements are available to help in the diagnosis and evaluation of pulmonary illness, to assess the necessity of treatment, and to gauge how well respiratory care is working.^[18]

The following categories are included in pulmonary function tests:^[18]

- 1) Dynamic flow rates of gases through the airways. (Pulmonary mechanics)
Lung volumes and capacities
- 2) Diffusion capacity

The purpose of pulmonary mechanics is to evaluate the lungs' capacity to swiftly transfer large amounts of air through the airways in order to detect airway obstruction. This test is most frequently administered using the forced expiratory vital capacity (FVC) manoeuvre. wherein the individual takes their full breath and then exhales as quickly and thoroughly as they can.^[8]

The two methods of recording the FVC test are: -^[8]

- Volume-time curve-The most common measurements made from this curve are the force expiratory volume in 1 second (FEV1).
- Flow-volume curve-The most common measurement from this curve is peak expiratory flow (PEF)

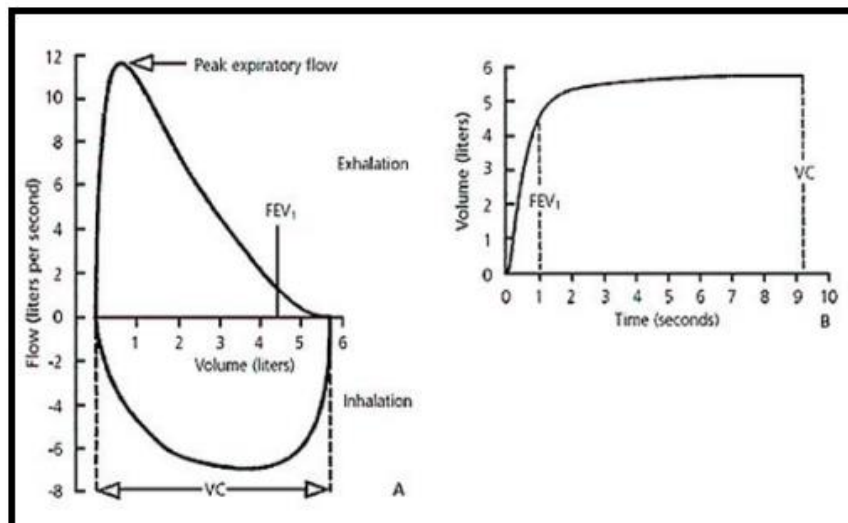


Fig 5: VOLUME TIME AND FLOW TIME CURVE LUNG VOLUMES AND CAPACITIES

➤ **LUNG VOLUMES AND CAPACITIES:**

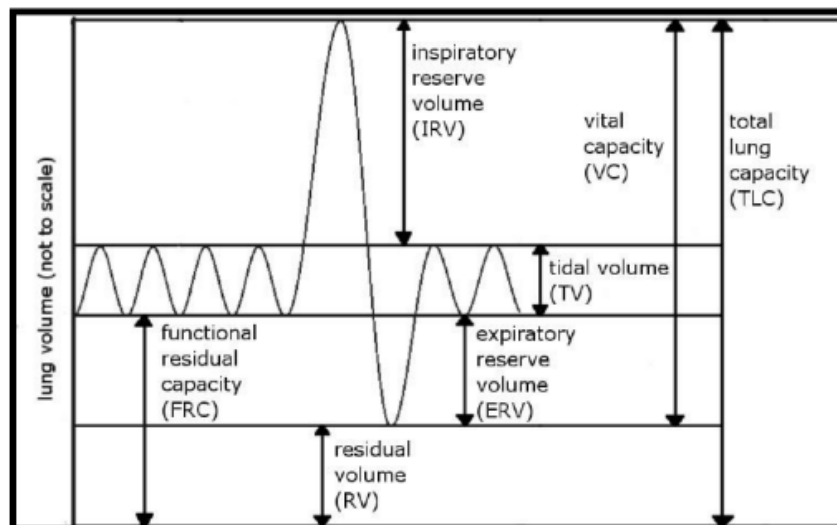


Fig 6: LUNG VOLUMES AND CAPACITIES

The lung volumes: [8]

- 1) Tidal volume (TV) is the volume of air inhaled & exhaled at each breath during normal quiet breathing
- 2) Inspiratory reserve volume (IRV) is the maximal amount of air that can be inhaled from the end of a normal inspiration.

3) Expiratory reserve volume (ERV) is the maximal amount of air that can be expired after a normal exhalation

4) Residual volume (RV) is the volume of gas that remains in the lungs at the end of a maximal expiration. Change in residual volume can help in the diagnosis of certain medical condition. Residual volume (RV) is increased in obstructive lung condition and it is decreased in restrictive lung disease

The capacities include: - [8]

1) **Total lung capacity (TLC)** is the amount of gas the lung contains at the end of a maximum inspiration. It is made up of all four lung volumes. An increase in total lung capacity is seen with hyperinflation such as emphysema. A decrease in TLC may be seen in restrictive lung disease.

2) **Vital capacity (VC)** is the maximum amount of gas that can be expired from the lungs by forceful effort following a maximum inspiration. It contains the IRV, TV & ERV.

3) **Inspiratory capacity (IC)** is the maximum amount of air that can be inspired from the resting expiratory level.

4) **Functional residual capacity (FRC)** is the volume of air remaining in the lungs at the resting expiratory level. It contains the ERV and the RV. FRC prevents large fluctuation in PaO₂ with each breath. An increase in FRC represents hyperinflation of the lungs. It causes the thorax to be larger than normal which results in muscular inefficiency and some mechanical disadvantage.

Diffusion capacity (DLCO): - [8] is the ability of the lungs to transfer gases across the alveolar-capillary membrane. Normal values for the DLCO using the single breath technique for a young healthy man is 40 ml/min/mmHg.

E.1) - Bartolome R. Celli (2000) “The importance of spirometry in COPD and asthma” by, Spirometry confirms the diagnosis, which was indicated by the

history and physical examination and was characterized by a low FEV1 level that was not responsive to bronchodilators. Following a diagnosis, there is no commonly used stage or severity rating system. COPD now uses a single metric, such as FEV1, to grade the condition, which has a limited function in managing the condition, in contrast to asthma. The inflammatory condition known as asthma is likewise characterized by restricted airflow. However, unlike COPD, the airflow limitation is very reversible, either on its own or with treatment. Results have improved after multiple lung function assessments with portable peak flowmeters. Consequently, it is advised that routine flow determination be done frequently. control of asthma. Anti-inflammatory medication treatment and careful observation of lung function ought to contribute to a reduction in the morbidity and mortality linked to asthma.^[34]

E.2) T R Schermer, J E Jacobs, et al “Validity of spirometric testing in a general practice population of patients with chronic obstructive pulmonary disease (COPD)” (2003) conducted a study to investigate the validity of Spirometric tests performed in general practice: In order to compare Spirometric testing with the gold standard (Spirometric tests conducted in a pulmonary function laboratory), a repeat within-subject comparison was conducted in 388 patients with 61 general practices and four labs with chronic obstructive pulmonary disease (COPD). A spirometry training program was completed by general practitioners and practice assistants. Practice minus laboratory value was used to calculate the differences between the laboratory and general practice tests for the subjects' forced expiratory volume in one second and forced vital capacity (FEV1 and FVC). The percentage of tests with 200 ml or 5% FEV1 repeatability was used as a quality indicator. Relevant Spirometric indicators assessed by qualified general practice personnel were found to be slightly but statistically substantially higher than those assessed in pulmonary function labs.^[35]

D.3) Timothy J. Barreiro et al “An Approach to Interpreting Spirometry” (2004) Spirometry was an effective method for identifying, monitoring, and treating lung conditions. Spirometry's dependability has significantly enhanced because to modern technology, which has also made it quite simple to incorporate

into a routine office visit. Important components of test quality include patient involvement and effort as well as the interpreter's knowledge of appropriate reference values. Bronchodilator challenge testing, carbon monoxide diffusing capacity, and static lung volumes are some of the additional tests needed for some patients. Only pulmonary function laboratories have the more sophisticated equipment and expertise needed for these tests, which can further characterize lung processes. [36]

D.4), Stuart W. Stoloff et al (2010) "Diagnosis and treatment of patients with chronic obstructive pulmonary disease in the primary care setting: focus on the role of spirometry and bronchodilator reversibility" To make a conclusive diagnosis of COPD, spirometry was needed; handheld spirometers were created as practical and affordable substitutes for conventional spirometers and could make spirometry easier to perform in Primary Care. Despite the fact that absence of complete reversibility was a component of COPD, numerous patients have shown a significant response to bronchodilators. It is noteworthy that patients may exhibit a considerable volume response (e.g., IC, FVC) even in the lack of a flow response (FEV1). In patients with COPD, bronchodilator medication has been demonstrated to improve exercise tolerance and reduce hyperinflation, two crucial therapeutic objectives. Since primary care doctors are often the first to see patients, it is imperative that they identify those who are at risk for COPD, utilize spirometry to make an accurate diagnosis, and treat those patients appropriately. [37]

[F] SIX MINUTE WALK TEST:

The 6-min walking test (6MWT) test has gained importance in the assessment of functional exercise capacity in patients with chronic -obstructive pulmonary disease (COPD). The test evaluates the global and integrated responses of the pulmonary, cardiovascular and muscular component and reflects the functional exercise level for daily physical activities. The 6MWT is a submaximal exercise test, but-it correlates-well with the maximal cardiopulmonary exercise test and appears to be better tolerated and more reflective of the activities daily living than other walk tests.

F.1), C. Casanova et al, May 30 2006 studied, “Distance and Oxygen Desaturation During the 6-min Walk Test as Predictors of Long-term Mortality in Patients with COPD” and concluded that the 6-min walking distance test provides increasingly useful information as the severity of chronic obstructive pulmonary disease increases. The 6-min walking test (6MWT) test is used in clinical practice and research into patients with chronic obstructive pulmonary disease (COPD). However, little is known about natural long-term change in this parameter. The 6MWT was measured at baseline and then annually for 5 yrs in 294 patients with COPD and its annual rate of decline was determined. Forced expiratory volume in one second (FEV_1) was also measured and the relationship between changes in both markers was explored. At baseline, the median 6MWT was 380 m (range 160–600 m). It declined by 19% ($16 \text{ m}\cdot\text{yr}^{-1}$) over the 5 yrs compared with baseline in patients with American Thoracic Society/European Respiratory Society stage III COPD (FEV_1 30–50% predicted) and by 26% ($15 \text{ m}\cdot\text{yr}^{-1}$) in patients with stage IV COPD ($FEV_1 < 30\%$ pred) Over the 5-yr follow-up, the proportion of patients with a minimal clinically significant decline of 54 m increased with the severity of the disease. It was 24% in stage II, 45% in stage III, and 63% in stage IV disease. In contrast, the rate of decline of FEV_1 was greater in patients with milder airflow obstruction and lesser in patients with lower absolute FEV_1 values.^[38]

F.2), Imen Ben Cheikh Rejbi1, et al 29 June 2010, studied “Changes in six-minute walking distance during pulmonary rehabilitation in patients with COPD and in healthy subjects” and concluded in the study of changes in six-minute walk test during pulmonary rehabilitation in COPD, shows the average walking distance of both groups increased significantly during the first week of training ($P, 0.05$), and within two weeks all the subjects showed an increase in 6MWT. Patients completed two 6MWT tests per evaluation of ≥ 30 min apart, following a modified protocol. The modifications included the use of a 36 m long corridor, encouragement of the patient, oxygen saturation monitoring and oxygen provision to the patients whose oxygen saturation decreased to $< 85\%$. The respiratory therapist carried the oxygen tank. The longest of the two walk distances was used

in analysis. The same respiratory nurse on each site performed the evaluations during the time span of the study. For all the patients, the minimal time between two evaluations was 1 yr. This minimised the possible influence of pulmonary rehabilitation on the walked distance in the nine patients who enrolled in pulmonary rehabilitation.^[39]

[G] CHEST EXPANSION:

Chest expansion, defined as the difference in thoracic girth after maximum inspiration and maximum expiration, is one indicator of chest wall mobility. Normal chest wall mobility is improvement for effective lung expansion and subsequent ventilation. Chest wall mobility is often assessed by measuring chest expansion with inch tape (measure tape), Thoracic expansion equals thoracic circumference at the end of forced inspiration minus circumference at the end of forced expiration. Chest excursion measurements are used to evaluate a patient's baseline status, treatment effectiveness, and progression of disease with regards to chest wall mobility and respiratory muscle function 1 which is measured at axilla, nipple and xyphi sternum. Generally, chest expansion should be roughly 2 to 5 inches (5 to 12 cm) for most adults, although this range can vary based on factors such as age, sex, and overall health.

G.1) Australian Government Department of Health and Ageing, 15 July 2004

Chest expansion should be measured by a uniform method. Using a tape measure the change in circumference of the patient's chest at the level of the 4 intercostal space is recorded.^[40]

G.2) Rufus Adesoji Adedoyin et al did study “Reference Values for Chest Expansion among Adult Residents in Ile-Ife” (2012) The findings indicated that male participants had a significantly larger chest than female participants in the upper thoracic region (2.6+1.4 cm, 2.2+1.2 cm) and lower thoracic region (2.3+1.2 cm, 1.7+1.1 cm) ($p < 0.05$) respectively. Male and female participants' chest expansions at both .The upper and lower thoracic peaks between the ages of 20 and 29, and then declines as people age. The findings indicated that, for both

upper and lower thoracic regions, chest expansion was adversely connected with age in males ($r=-0.370$; $p<0.05$); $r=-0.153$; $p<0.05$); and females ($r=-0.319$; $p<0.05$); $r=-0.458$; $p<0.05$). There was a significant correlation between female participants' BMI and chest expansion ($r=-0.141$, 0.197 ; $p<0.05$). Values below the 25th, between the 25th and 75th, and above the 75th are regarded as poor, moderate, and good excursions, respectively, for all participants. The numbers are 1.4 cm, 3.1 cm, 5.1 cm for the top expansion, and 1.0 cm, 2.7 cm, and 4.3 cm for the lower expansion.^[41]

MATERIALS
AND
METHODOGY

MATERIALS & METHODOLOGY

ETHICAL APPROVAL:

Ethical approval was obtained from Institutional Research and Ethics Committee prior to conducting the study.

• STUDY DESIGN:

An Experimental Study

• STUDY SETTING:

This study was conducted in different hospitals of Ahmedabad

• SAMPLE SELECTION:

The sample size consisted of 27 patients who were diagnosed with chronic obstructive pulmonary disease (GOLD CLASSIFICATION – GRADE – I & II) with help of digital spirometer.

• SAMPLE DESIGN:

Convenient sampling

• STUDY DURATION:

The study duration was 6 months.

• SAMPLE SIZE:

27 Male patients calculated from pilot study

SAMPLE SIZE ESTIMATION

$$n = 2(\sigma/\Delta)^2 (Z\alpha + Z1-\beta)^2$$

SELECTION CRITERIA: -

➤ INCLUSION CRITERIA:

- Patients who are diagnosed with COPD (ACCORDING TO GOLD CLASSIFICATION GRADE I & II)
- Males with age group between 40-60 years.
- Smokers, Non-smokers, Ex-smokers
- Patients who were willing to participate

➤ EXCLUSION CRITERIA:

- Patients with unstable vital parameters.
- Patients with any pulmonary conditions other than COPD.
- Patients with cardio-pulmonary surgeries.
- Patients with cardiac condition except Hypertension.
- Patients with any skin conditions and infections.
- Patients with neuro-psychiatric conditions
- Patients who have received bronchodilator less than 6 hours prior to data collection.
- Patients with any systemic dysfunctions

➤ OUTCOME MEASURES:

PRIMARY OUTCOME:

- Pulmonary functions (FEV₁, FVC, FEV₁/FVC, PEFr) By digital spirometry
- Six-minute Walk Test (6MWT)
- Middle chest expansion (MCE)

SECONDARY OUTCOME:

- SPO₂
- Respiratory rate (RR)
- Heart rate (HR)
- Blood pressure (BP)

➤ MATERIALS USED IN THE STUDY:

- Pen, Pencil, Paper
- Consent Form & Data collection sheet
- Sphygmomanometer & Stethoscope
- Plinth & chair
- Measuring tap
- Watch
- Pulse oximeter
- Kinesio Tape

➤ INSTRUMENTS USED IN THE STUDY:

- Digital spirometer
- Kinesio Tape



Fig 7: PFT MACHINE

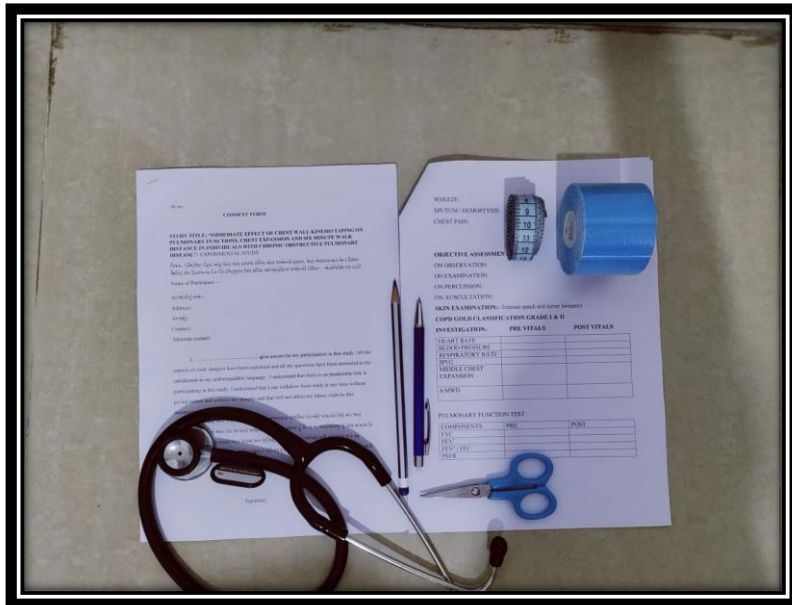


Fig 8.: CONSENT FORM AND KINESIO TAPE



Fig 8.1: SPHYGMOMANOMETER AND PULSE OXIMETER

PROCEDURE

Prior to undertaking the original study, a pilot study was conducted to find out the sample size of the study. Sample size was calculated using formula:

$n = 2(\sigma/\Delta)^2 (Z\alpha + Z1-\beta)^2$, and the 12 COPD patients were taken for the study.

Ethical approval was taken from Institutional Ethical Committee. Subjects from outpatient department of different hospitals in Ahmedabad were screened for the study. The whole procedure of the study was explained to the individuals and if they were willing to participate in the study they were asked to sign the consent form. Written consent informed was taken. 27 subjects were chosen for data collection. Subjects who were found according to inclusion and exclusion criteria were selected and assigned for the study.

On first visit a complete assessment was done, 6MWT and vitals (SPO₂, RR, HR, and BP) were taken. Skin patch test was done to rule out the skin allergy and hypersensitivity. Pre evaluation and Kinesio taping was done after at least 2 hours of meal and 6 hours of bronchodilators. The atmosphere of the room in which evaluation and Kinesio Taping session was done was soothing, quiet and well ventilated.

Pre-intervention evaluation from spirometry according to GOLD classification [Grade-1 & I1 COPD] was done and after the 20 min rest six-minute walk distance and middle chest expansion were measured. All participants then received Kinesio Taping rested for 1 hour. Then post evaluation and outcome measures were measured again after the Kinesio Taping.

➤ **MEASURING PULMONARY FUNCTION TEST:**

Before the measuring pulmonary functions of individuals, the Calibration of spirometer was done. The spirometry procedure was carried out as per the American Thoracic Society of standardization of spirometry guidelines. The procedure was explained to the subjects. Test was performed with the subjects in the sitting position with the nose clip. The test was repeated 3 times after adequate rest of 5 min to avoid the exertion. Average of all three repetitions is taken into consideration. These parameters included FVC, FEV₁, FEV₁/FVC, and PEFR.



Fig 10: PERFORMING PULMONARY FUNCTION TEST

➤ **MEASURING 6 MINUTE WALK TEST:**

The participants were explained about the whole procedure of six-minute walk test. Test was performed under the American thoracic society guidelines for six-minute walk test. A 30-meter hallway was marked. Participants were asked to take 15 to 20 minutes rest before starting the test. Then after participants were asked to walk on a corridor for 6 minutes. During the test performed if the participants feel any discomfort like dyspnoea or any other problem, immediately let the therapist know about that problem and they can take rest in between and again continue the test within the period of 6 minute. Vitals were measured continuously. The distance covered in the duration of 6 minutes was measured.

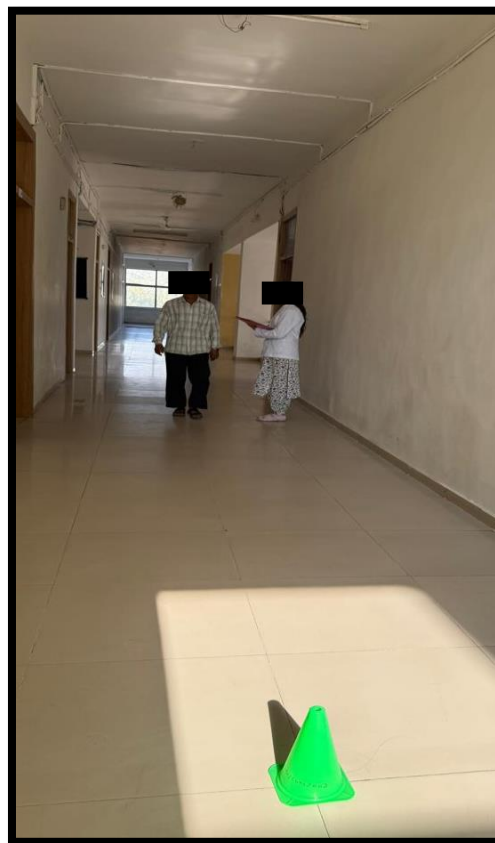


Fig 10: - MEASURING 6 MINUTE WALK DISTANCE

➤ **MEASURING MIDDLE CHEST EXPANSION:**

The chest expansion was taken as middle thoracic circumference at the nipple level with the measure tape.



Fig 11: - MEASURING MIDDLE CHEST EXPANSION

➤ **MEASURING SECONDARY OUTCOME:**

Measuring vitals (SPO₂, RR, HR, BP), participants were asked to sit comfortably and relax for 10 minutes in order to avoid any change in vitals.



Fig 12: - MEASURING SECONDARY OUTCOME

APPLICATION OF KINESIO TAPING

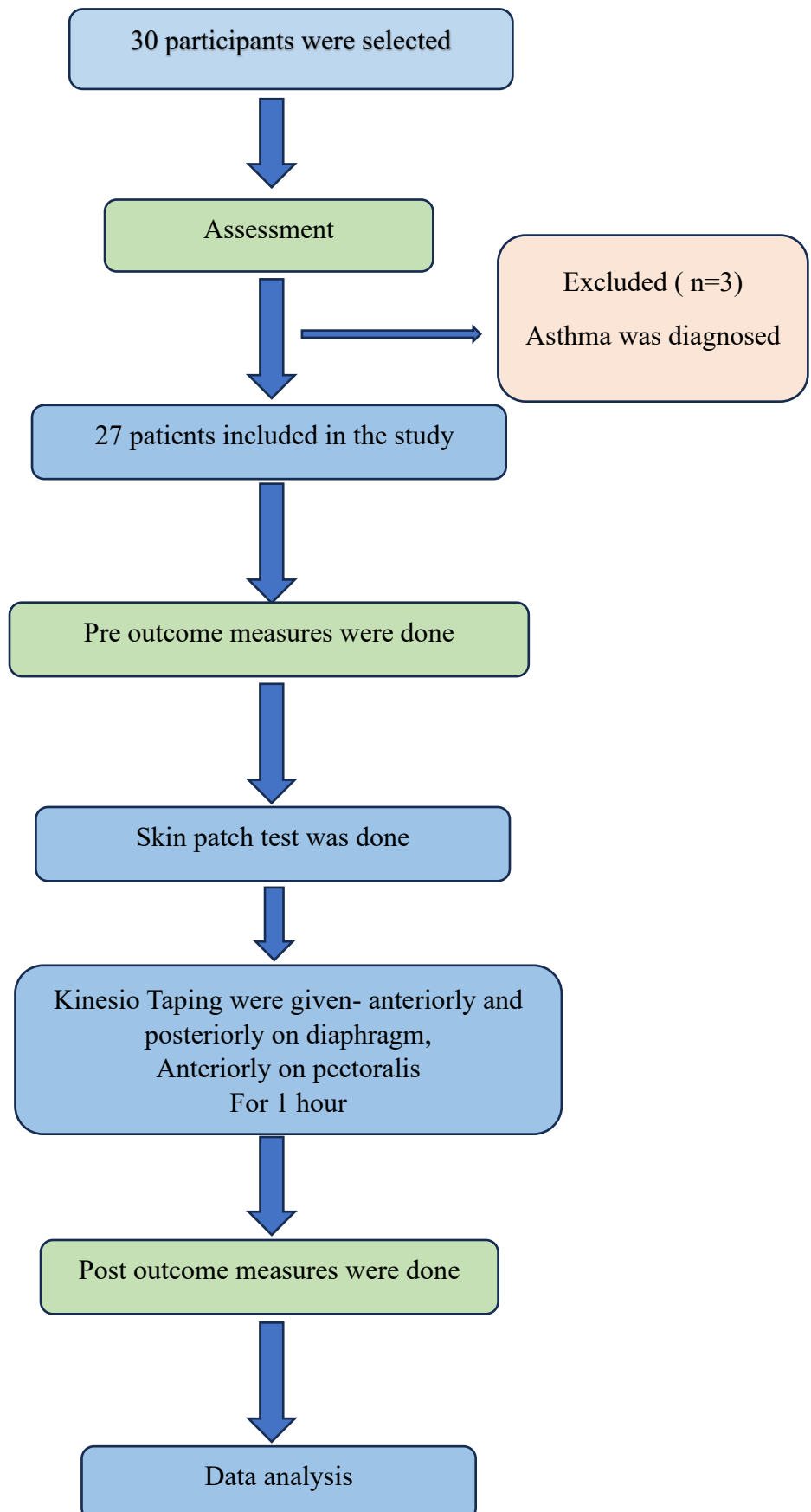
KT was applied anteriorly and posteriorly to facilitate respiratory muscles (primarily diaphragm)

KT was applied in an inhibitory method over pectoralis



Fig 13: - APPLICATION OF KINESIO TAPING

FLOWCHART



RESULT

RESULT

This study was performed to see the immediate effect of Kinesio taping on pulmonary functions and six-minute walk distance in COPD patients. In this study 27 subjects in age group of 40-60 years were taken for intervention. Pulmonary functions (FEV₁, FVC, FEV₁/FVC), 6MWD, Middle chest expansion, SPO₂, RR, HR, MBP were checked as the outcome measures. The statistical analysis was done in Microsoft Excel 2016 and IBM statistical package for social sciences (SPSS) Version20.0.

The results are expressed in mean, standard deviation and percentage.

TABLE 1: DEMOGRAPHIC DETAILS OF THE PARTICIPANTS

No. Of Participants	27
AGE	54.19 ±4.76
BMI	24.01±1.32

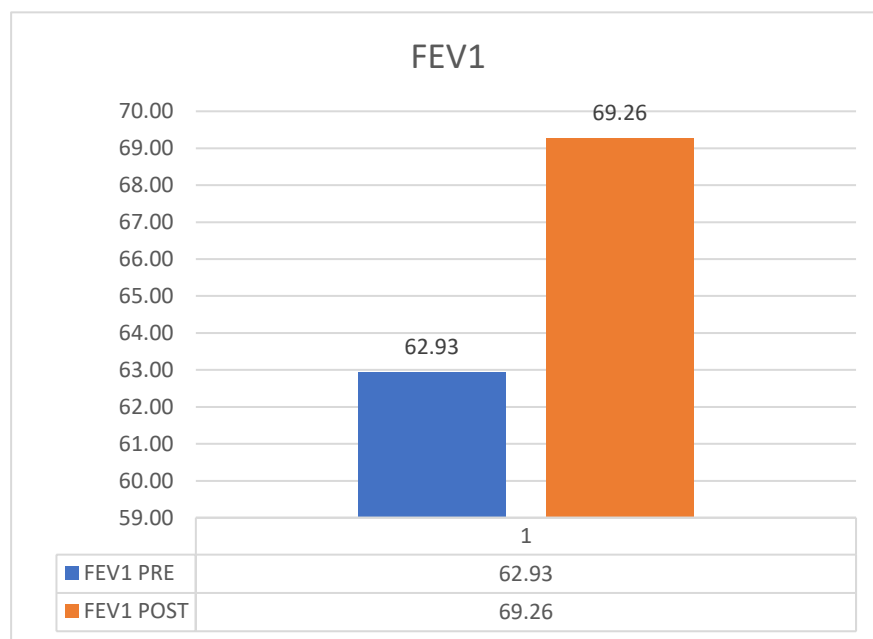
TABLE 2: STATISTICAL TEST FOR OUTCOME MEASURES

OUTCOME MEASURES	TEST USED TO ANALYSIS IN COPD GROUP
FEV ₁	PAIRED T-TEST
FVC	PAIRED T-TEST
FEV1/FVC	WILCOXON SIGNED RANK T-TEST
PEFR	PAIRED T-TEST
6MWD	PAIRED T-TEST
MCE	WILCOXON SIGNED RANK T-TEST
SPO ₂	PAIRED T-TEST
RESPIRATORY RATE	WILCOXON SIGNED RANK T-TEST
HEART RATE	PAIRED T-TEST
MBP	PAIRED T-TEST

TABLE 3: EFFECT OF KINESIO TAPING ON FEV₁

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
FEV ₁ [L]	62.9±5.51	69.2±5.20	-27.12	0.000

Paired T-test was performed to analysis the effect of Kinesio taping on FEV₁. Here the “p” value is **<0.05** which is statistically significant.

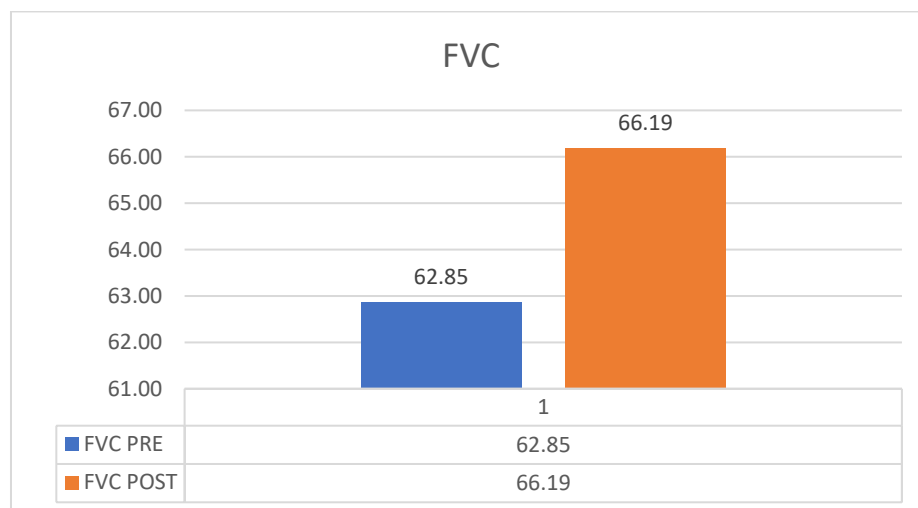


GRAPH 1: PRE AND POST EFFECT OF FEV₁

TABLE 4: EFFECT OF KINESIO TAPING ON FVC

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
FVC [L]	63.0±7.75	66.3±7.77	-3.594	0.001

Paired T-test was performed to analysis the effect of Kinesio taping on FVC. Here the “p” value is <0.05 which is statistically significant.



GRAPH 2: PRE AND POST EFFECT OF FVC

TABLE 5: EFFECT OF KINESIO TAPING ON FEV₁/FVC

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
FEV ₁ /FVC [%]	59.2±9.18	64.0±7.88	-5.885	0.000

Wilcoxon test was performed to analysis the effect of Kinesio taping on FEV₁/FVC. Here the “p” value is <0.05 which is statistically significant.

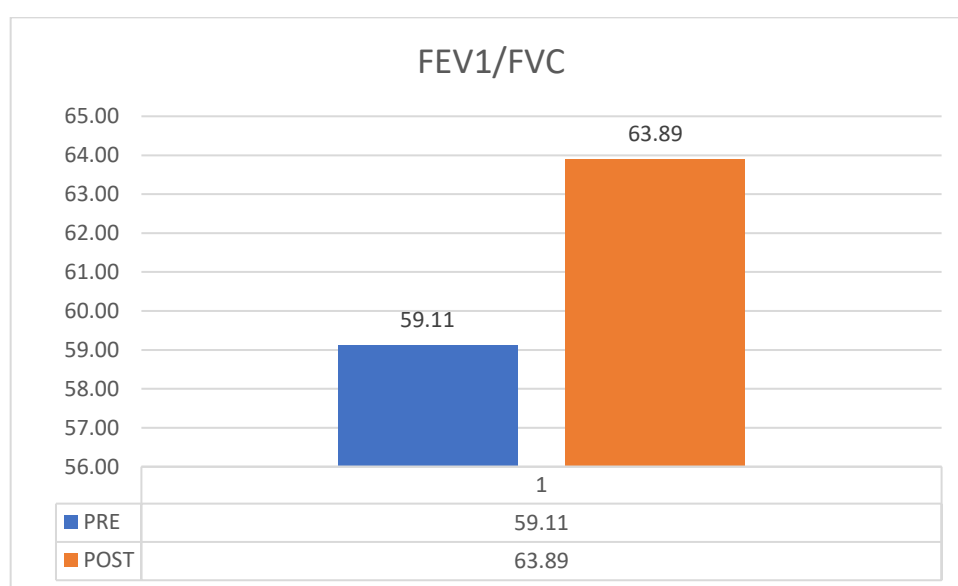
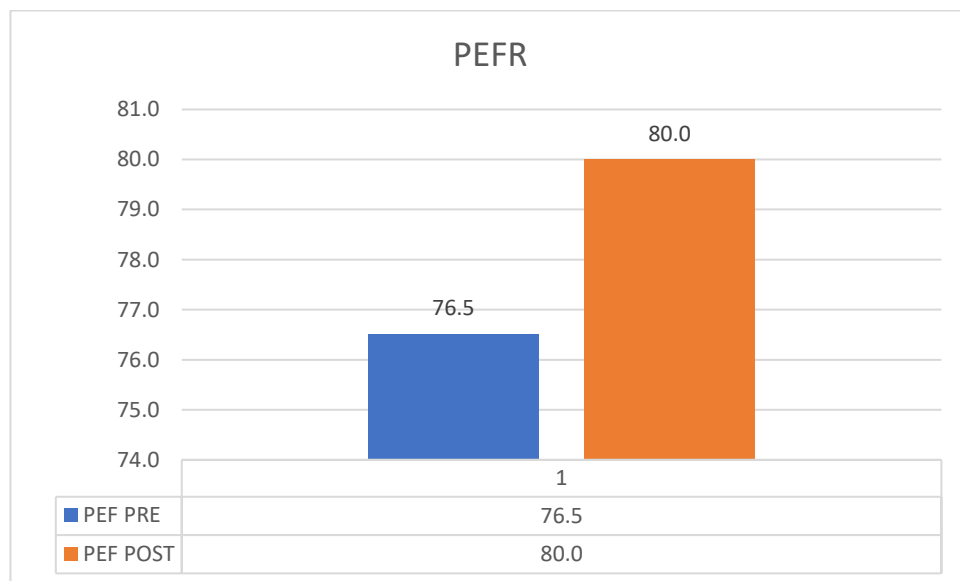
**GRAPH 3: PRE AND POST EFFECT OF FEV₁/FVC**

TABLE 6: EFFECT OF KINESIO TAPING ON PEFR

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
PEFR [L/s]	76.8±19	80.5±20	-2.870	0.008

Paired T-test was performed to analysis the effect of Kinesio taping on PEFR. Here the “p” value is >0.05 which is statistically not significant.



GRAPH 4: PRE AND POST EFFECT OF PEFR

TABLE 7: EFFECT OF KINESIO TAPING ON 6MWD

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
6MWD [m]	361.6±40	379±36	-8.307	.000

Paired T-test was performed to analysis the effect of Kinesio taping on 6MWD.

Here the “p” value is <0.05 which is statistically significant.

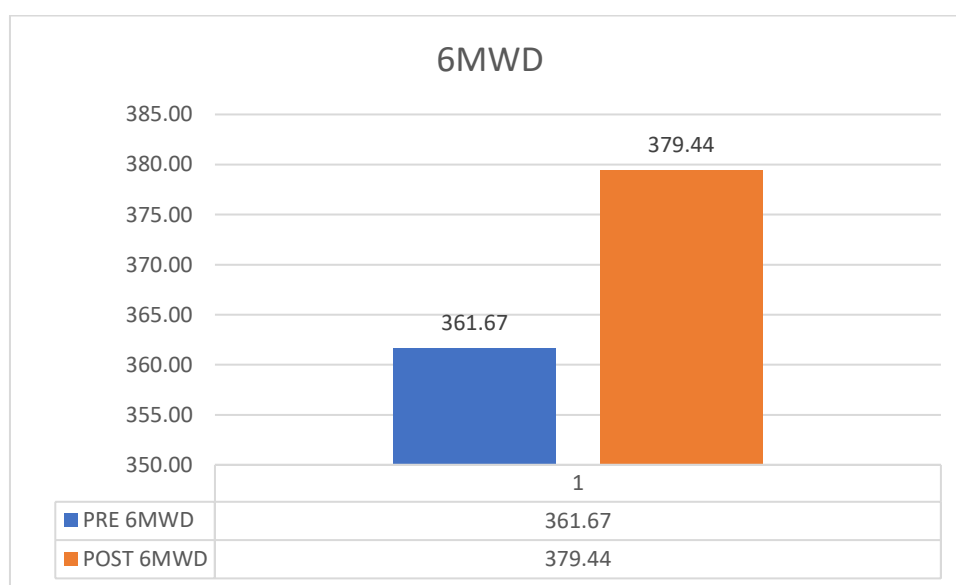
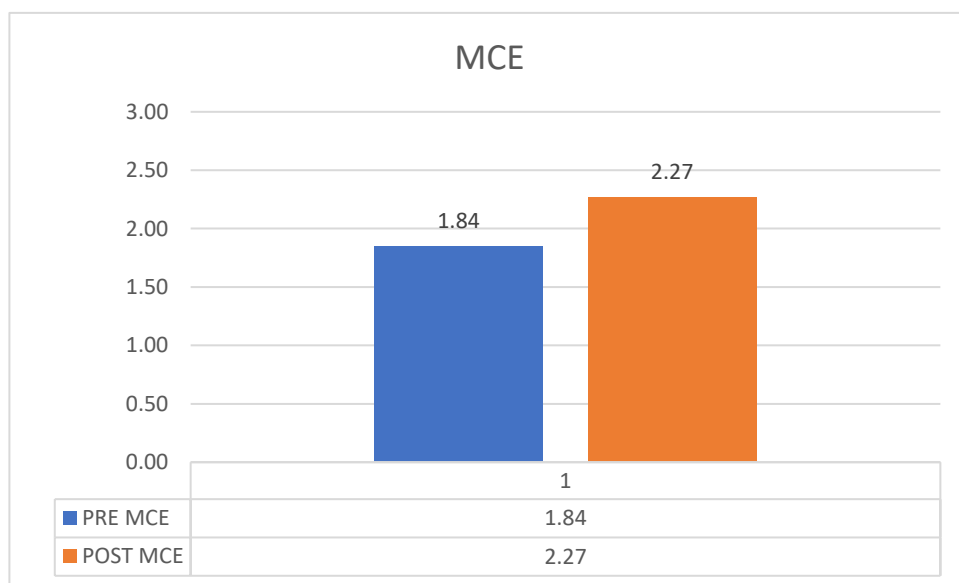
**GRAPH 5: PRE AND POST EFFECT OF 6MWD**

TABLE 8: EFFECT OF KINESIO TAPING ON MCE

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	z- VALUE	p-VALUE
MCE[Inch]	1.95±0.79	2.34±0.84	-4.125 ^b	.004

Wilcoxon test was performed to analysis the effect of Kinesio taping on MCE.
Here the “p” value is <0.05 which is statistically significant.



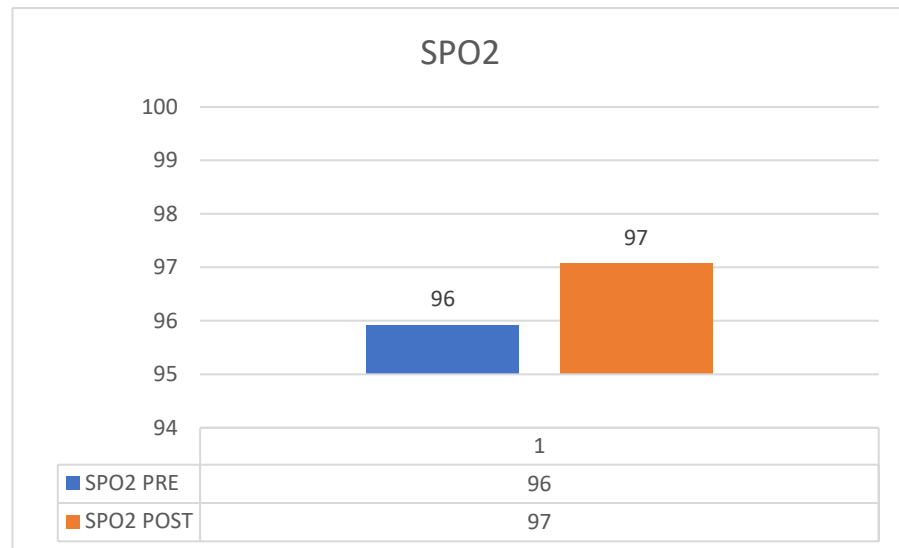
GRAPH 6: PRE AND POST EFFECT OF MCE

TABLE 9: EFFECT OF KINESIO TAPING ON SPO₂

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
SPO ₂ [%]	95.93±1.12	97.0±1.05	-6.905	0.000

Paired T-test was performed to analysis the effect of Kinesio taping on SPO₂

Here the “p” value is <0.05 which is statistically significant.



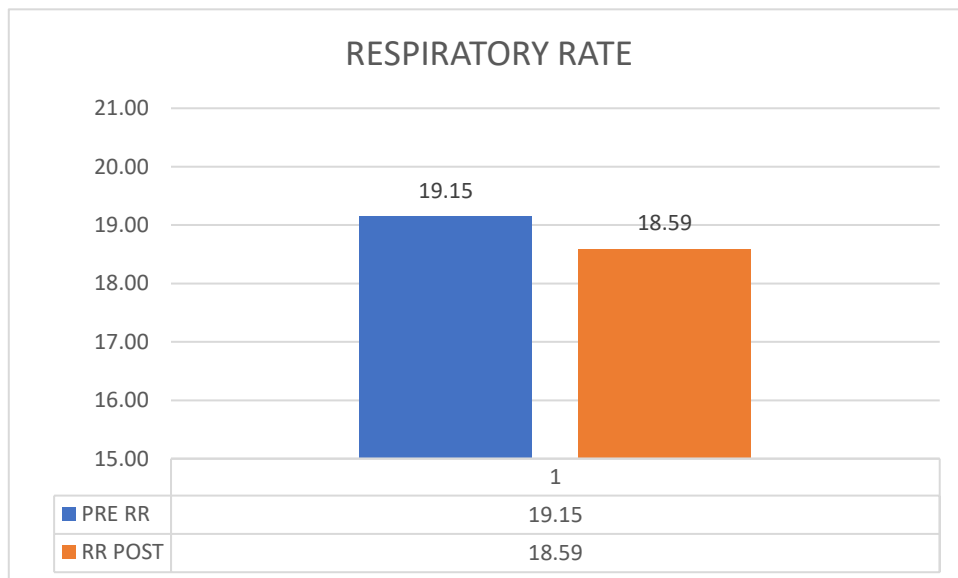
GRAPH 7: PRE AND POST EFFECT OF SPO₂

TABLE 10: EFFECT OF KINESIO TAPING ON RESPIRATORY RATE

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	z- VALUE	p-VALUE
RR [rpm]	19.1±1.04	18.5±1.060	-3000 ^b	0.003

Wilcoxon test was performed to analysis the effect of Kinesio taping on Respiratory Rate.

Here the “p” value is <0.05 which is statistically significant.

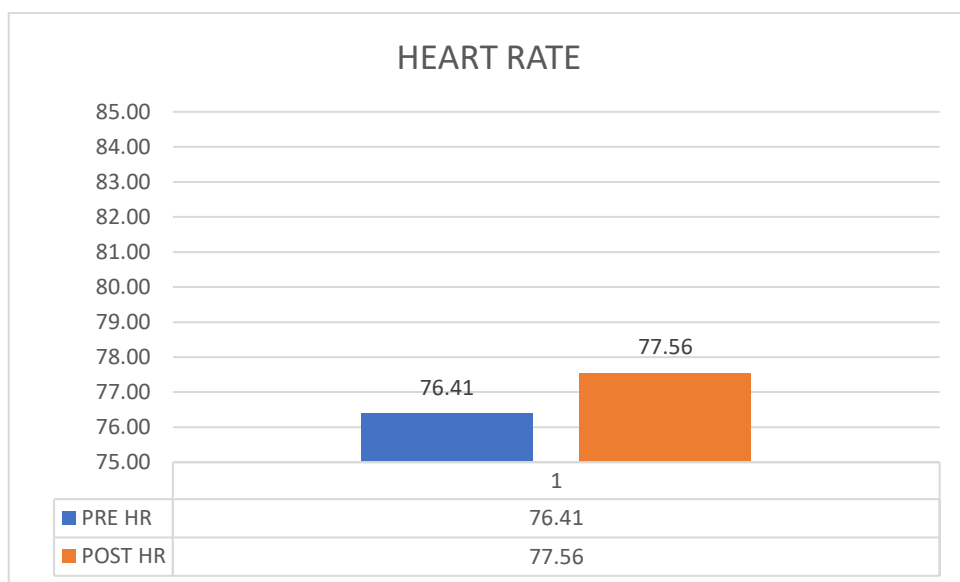


GRAPH 8: PRE AND POST EFFECT OF RR

TABLE 11: EFFECT OF KINESIO TAPING ON HEART RATE

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
HR [bpm]	76.4±5.59	77.56±4.85	-2.807	0.009

Paired T-test was performed to analysis the effect of Kinesio taping on HR.
Here the “p” value is >0.05 which is statistically not significant

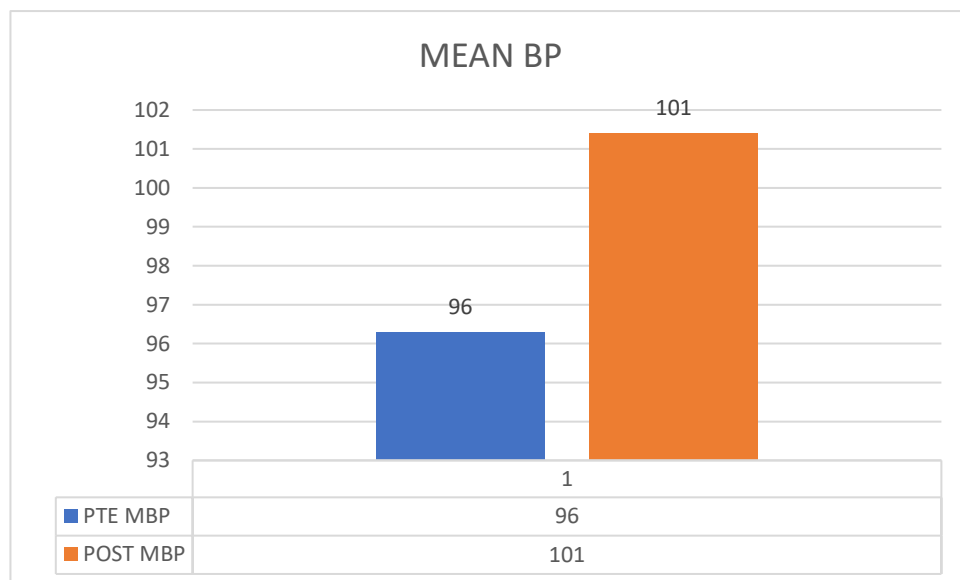


GRAPH 9: PRE AND POST EFFECT OF HEART RATE

TABLE 12: EFFECT OF KINESIO TAPING ON MEAN BP

OUT COME	PRE (MEAN ±SD)	POST (MEAN ±SD)	t- VALUE	p-VALUE
MBP [mmHg]	96.0±10.9	101±8.1	-3.851	0.069

Paired T-test was performed to analysis the effect of Kinesio taping on BP.
Here the “p” value is >0.05 which is statistically not significant



GRAPH 10: PRE AND POST EFFECT OF MEAN BLOOD PRESSURE

DISCUSSION

DISCUSSION

This study was conducted to find the immediate effect of Kinesio taping on pulmonary functions, six-minute walk distance and middle chest expansion in chronic obstructive pulmonary disease (COPD). An interventional study was conducted on 27 individuals ageing between 40-60 years having Grade I & II stable chronic obstructive pulmonary disease assessed by digital spirometry according to GOLD CLASSIFICATION. Kinesio taping were given where pre and post outcome measured with the help of Spirometry, six-minute walk test, middle chest expansion, and with secondary outcome SPO₂, Respiratory Rate, Heart Rate, Blood Pressure.

The statistical analysis was done in Microsoft Excel 2016 and IBM statistical package for social sciences (SPSS) Version 20.0. Comparable mean score of pre and post outcome measures were tested for normality. Thus, statistical analysis was done by using non-parametric test (Wilcoxon sign rank test) for, MCE, RR.

Kinesio taping proved significant statistical improvement in FEV₁, FVC, FEV₁/FVC, SPO₂, 6MWT and RR in COPD.

Skeletal muscle deterioration in patients with COPD has a direct impact on functional exercise capacity.^[42,43] Because respiratory muscles are anatomically similar to striated muscles, they become exhausted after prolonged use. The continuing weariness generates inhibitor signals and reduces motor commands, resulting in respiratory insufficiency in COPD patients.^[44] Dynamic hyperinflation is also a primary cause of functional exercise restriction in COPD.^[45] Increased lung volume shortens the length of the inspiratory muscles and produces functional weakening. The diaphragm's effect on the rib cage is particularly adversely affected.^[46]

This study found that applying KT to the diaphragm muscle improves the performance of FEV₁, FVC, and FEV₁/FVC in people with COPD.. A method of stimulation was employed in present study to enhance the activity of the respiratory muscles. By stimulating the diaphragm muscle's fibres and activating the skin's mechanoreceptors that affect the diaphragm through the fascia, KT is a Muscle

Facilitation treatment. Kinesiology taping is the primary method of cutaneous sensorimotor stimulation of the diaphragm, which also helps in retraining to function well. This result is supported by the similar study done by **Burcu Metin Okmen et al** 2019 in Turkey. According to it, Kinesio taping improved in the FEV1 and FEV1/FVC levels in spirometric tests in patients with COPD ^[55]

PEFR shows no improvement after applying KT.. Given that it needs enough time to improve, the brief duration may be the cause of the lack of noticeable impact. The primary reasons of low PEFR include narrowing of the airways, decreased muscle strength, decreased lung elastic recoil, and limited movement of the chest wall.^[47] The authors believed that a single KT session would not result in a substantial change in the pulmonary function of these subjects. A study by **HANY E. OBAYA et al. (2022)** in Mansoura also examined the effects of KT treatment on pulmonary function and discovered no appreciable improvement in PEFR values after KT administration in COPD patients ^[56]

There was a significant improvement in 6 minute walk distance in COPD patients after application of KT, suggesting that tapping may enhance functional exercise capacity. This improvement is likely due to improved chest wall mechanics, increased respiratory function, decreased muscle fatigue, and better posture. These factors combine to increase a person's capacity for physical activity, as demonstrated by the six-minute walk test. In Pune, **Gaurang Baxi et al. (2020)** investigated Kinesio tape as an adjuvant therapy for individuals with COPD and found that it significantly enhanced O₂ saturation and reduced dyspnoea in the 6-minute walk test. As a result, the patients were able to move more easily. ^[47]

The improvement in middle chest expansion followed by the Kinesio Tape. which might be due to flexible system provided by KT, that promotes movement of the chest wall. As a consequence, the thoracic cage becomes more movable, allowing for a greater expansion during inhalation. maximize the posture and function of the intercostal muscles, which are crucial for the development of the rib cage. Which was supported by the Thailand study by **Tadsawiya Padkao et al (2020)**, implies the contraction of the inspiratory muscles most notably the diaphragm and external intercostal muscles produces the force contact between the chest wall and the expansion

of the chest wall and lungs. As a result, the pleural pressure decreases, creating a pressure gradient that allows air to enter the lungs.^[57]

After application of KT there was improvement in mechanics of chest wall, which helps lung to expand fully during expiration, which could be the reason behind improvement on oxygen saturation. KT increases expansion which causes more air to enter the alveoli, where carbon dioxide and oxygen are exchanged. In agreement of result, the similar study done by **Ganesh B. R et al** in Karnataka: The tape was applied across the thorax's outer margins, following its natural movements during inhaling, to achieve this. The elastic properties of the tape provide a counterbalancing effect as the person exhales, aiding in the expiration process. ^[33]

Respiratory Rate was not shown any improvement after application of KT. Kinesio taping gently supports the chest wall, and because of its increased mechanical efficiency, the breathing muscles can function more efficiently. Kinesio taping can encourage a more relaxed and effective breathing pattern by reducing the strain on the respiratory muscles. Because the body no longer needs to breathe quickly to meet its oxygen demands, this lowers the respiratory rate. The body can sustain efficient ventilation with less energy expenditure by minimizing the muscle fatigue that might arise from prolonged or laborious breathing.

Mean Blood Pressure was not shown any improvement after application of KT. Kinesio taping may improve lymphatic and blood flow, which could result in a higher venous return to the heart. MBP may be gently raised by a minor rise in cardiac output (CO) brought on by a higher venous return. On the other hand, **M Shah et al 2018** in UK stated that. Its mechanism of action may involve stimulating the muscles, which increases local blood circulation, or stimulating the muscles, fascial structures, and dermatological area with a type of passive massage might increase slightly if it applies the KT with longer duration.^[58]

Heart Rate was not shown any improvement after application of KT, because afferent sensory pathways may be activated by the sensory stimulation that kinesio taping gives to the skin and underlying muscles. A minor sympathetic nervous system activation brought on by this stimulus may cause a slight rise in heart rate. To sustain cardiac output, it might improve venous return, causing a slight compensatory rise in

heart rate. Moreover **Malehorn et al. (2013)** ^[51] in USA. Stated that this kind of ventilation increase happened without resulting in more oxygen being taken in. Therefore, the slow component of VO_2 and ventilatory muscles both have a role in exercise weariness. The KT might have lessened the pressure that the chest muscles had to work against or the contractile load on the respiratory muscles.^[51]

The other study by **Jan Szczegielniak January 2007:** - ^[53] in Kup. They examined on all patients with acute bronchial asthma by applying Kinesio Tex tape to improve efficiency, normalize extra breathing muscle tone, and correct inspiratory chest posture. Cough reflexes and secretions increased after one hour of Kinesio Tex treatment, making spirometry challenging. The patients' condition improved after 24 hours of treatment. All spirometric parameters improved in comparison to their original values. The capacity to cover even surface distance improved. As a result, the findings imply that Kinesio Taping can be beneficial in physiotherapy for individuals with acute asthma.

Even after giving participants with respiratory impairments a Kinesio taping session, some pulmonary functions (PEFR) did not significantly improve. Kinesio tape works on Myelinated cells which are bombarded with impulses when Kinesio taping triggers low threshold mechanoreceptors in muscle spindles, fascia, Golgi tendon organs, and skin. Nerve fibres regulate proprioceptive and kinaesthetic perception, as well as muscle tone and strength. Additionally, by providing mechanical support to the rib cage and accessory inspiratory muscles, kinesiology tape can help with dyspnoea.

.

CONCLUSION

CONCLUSION

Here according to the result of the study, the null hypothesis “There would be no immediate effect of Kinesio taping on pulmonary functions (FEV_1 , FVC, FEV_1/FVC), SPO_2 , 6MWT and RR in individuals with chronic obstructive pulmonary disease” is rejected and so the experimental hypothesis “There would be immediate effect of Kinesio taping on pulmonary functions (FEV_1 , FVC, FEV_1/FVC) SPO_2 , 6MWT and RR and in individuals with chronic obstructive pulmonary disease” is accepted.

It can be concluded from study that Kinesio taping can clinically improve the pulmonary functions (FEV_1 , FVC, FEV_1/FVC), six-minute walk test, middle chest expansion, RR and SPO_2 for short time in COPD, but there is no significant improvement in PEFR, HR and BP with the treatment of Kinesio taping.

CLINICAL IMPLICATION

CLINICAL IMPLICATION

The present study suggested that there is improvement in pulmonary functions (FEV_1 , FVC, FEV_1/FVC), RR, 6MWD, and saturation level (SPO_2) in chronic obstructive pulmonary disease.

Kinesio taping can be beneficial in treatment of relieving dyspnoea & acute excursion, improve lung functions, and improve muscle flexibility. So, in clinical practice Kinesio taping should be recommended for pulmonary rehabilitation in patient with COPD as non-pharmacological management along with conventional therapy.

**LIMITATION
AND
FUTURE
RECOMMENDATION**

LIMITATION

- Single session study & short-term effect.

FUTURE RECOMMENDATION

- Further study can also be done on other cardiopulmonary conditions.

SUMMARY

SUMMARY

TITLE: IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON PULMONARY FUNCTIONS AND SIX MINUTE WALK DISTANCE AND CHEST EXPANSION IN INDIVIDUALS WITH CHRONIC OBSTRUCTIVE PULMONARY DISEASE – AN INTERVENTIONAL STUDY.

BACKGROUND: COPD is characterised by airflow limitations that is not fully reversible and it may include diseases that causes airflow obstruction like emphysema, chronic bronchitis or a combination of the disorder. COPD caused by three primary symptoms- dyspnoea, Cough, sputum production.

As a management of COPD there are many physiotherapy techniques in pulmonary rehabilitation such as airway clearance techniques, respiratory muscle training, etc. And other few traditional medicine techniques that we can also use as the nonpharmacological management for medical conditions. Kinesio taping is one of the alternative traditional medicine techniques.

Kinesiology taping is a non-pharmacological method of treatment used in the prevention and rehabilitation of musculoskeletal-related conditions including management of pain. Kinesiology taping involves application of elastic, thin, non-medicated cotton based water-resistant adhesive tape to the skin using a variety of techniques. kinesiology taping technique involves stretching adhesive kinesiology tape and applying it to the skin overlying muscles of inspiration; it is claimed that this causes tactile stimulation. There is tentative evidence from small-scale clinical studies that kinesiology taping may improve inspiratory muscle strength and respiratory function, and alleviate breathlessness

AIM:

To investigate the immediate effect of chest wall Kinesio taping on pulmonary functions and six-minute walk distance and chest expansion in patient with chronic obstructive pulmonary disease.

MATERIALS & METHODOLOGY:

Ethical approval was obtained from Institutional Ethical Committee. Subjects from outpatient department of different hospitals in Ahmedabad were screened for the study. The whole procedure of the study was explained to the individuals and written consent informed was taken. 27 subjects were chosen for data collection. Subjects who were found according to inclusion and exclusion criteria were selected and randomly assigned for the study.

On first visit a complete assessment was done and vitals (SPO₂, RR, HR, and BP) were taken. Pre-intervention evaluation form spirometry according to GOLD classification [Grade-1 & II COPD] was done and after the 20 min rest six-minute walk distance and middle chest expansion were measured. All participants then received Kinesio taping. Post evaluation and outcome measures were measured again after the Kinesio taping.

RESULT: The result of this study showed significant statistical improvement in pulmonary function-FEV₁, PEFR, SPO₂, 6MWD and RR, with treatment of Kinesio taping.

CONCLUSION: It can be concluded from study that Kinesio taping can improve the pulmonary functions (specifically FEV₁, PEFR), and SPO₂ for short time in COPD.

KEYWORDS: Chronic Obstructive Pulmonary Disease, Kinesio Taping, Pulmonary Functions, Six Minute Walk Distance.

REFERENCES

REFERENCES

1. GOLD (2024) Global Initiative for Chronic Obstructive Lung Disease. Global Strategy for the diagnosis, management, and prevention of chronic obstructive pulmonary disease. Available from: <http://www.goldcopd.org>
2. Carolyn Kisner, Lynn Allen Colby "Therapeutic Exercise" Fifth edition, chapter-25, page-851-882
3. Ellen A Hillegass, H Steven Sadowsky "Essentials of cardiopulmonary physical therapy" Second edition, chapter-6 ,page-257-284
4. Ann Thomson, Alison Skinner, Joan Piercy "Tidy's Physiotherapy" Twelfth Edition, Chapter-13, Page-185-211
5. Susan B O'sullivan, Thomas J Schmitz "Physical Rehabilitation" Fifth edition, chapter-15, page-561-588
6. Robert L. Wilkins, James R. Dexter, Albert J. Heuer "Clinical Assessment in Respiratory care" Sixth Edition, Chapter-9, Page no-166-185
7. GOLD (2010) Global Initiative for Chronic Obstructive Lung Disease. Global Strategy for the diagnosis, management, and prevention of chronic obstructive pulmonary disease. Available from:
8. Robert E. Hyatt, Paul D. Scanlon, Masao Nakamura Interpretation of pulmonary Function tests-A practice Guide" Third edition, chapter-1, 2, 3,4. page-1-50
9. John E. Hodgkin, Bartolome R. Cell, Gerilynn L. Connors "Pulmonary rehabilitation- Guidelines to success"chapter-7, Page-83-94
10. Julia bott S Blumenthal, M Buxton, S Ellum, C Falconer, R Garrod, A Harvey,T Hughes, M Lincoln, C Mikelsons, C Potter, J Pryor, L Rimington, F Sinfield,C Thompson, P Vaughn, J Whit: Journal of British thoracic society Guidelinesfor the physiotherapy management of the adult, medical, spontaneously breathing patient." Thorax 2009; yolume-64 (suppl I) Page no-il -i50
11. Bianchi Roberto, Francesco Gigliotti, Barbara Lanini, Isabella Ramangnoli, Carla Castellani, Michela Grazzini and Giorgio Scana, Chest Journal, "Chest all

- kinematics and breathlessness during pursed lip breathing in patients with COPD" 2004.
12. Kase K. Clinical therapeutic applications of the Kinesio (! R) taping method. Albuquerque. 2003.
 13. Banerjee, Gourav & Johnson, Mark & Briggs, Michelle & Rose, Alison & Plant, Paul. (2021). Could kinesiology taping of the inspiratory muscles help manage chronic breathlessness? An opinion paper. *Progress in Palliative Care*. 29. 1-5. 10.1080/09699260.2021.1872137.
 14. Pamuk U, Yucesoy CA. MRI analyses show that kinesio taping affects much more than just the targeted superficial tissues and causes heterogeneous deformations within the whole limb. *J Biomech*. 2015 Dec 16;48(16):4262-70. doi: 10.1016/j.jbiomech.2015.10.036. Epub 2015 Oct 30. PMID: 26556717.
 15. Findley T, Chaudhry H, Dhar S. Transmission of muscle force to fascia during exercise. *J Bodyw Mov Ther*. 2015 Jan;19(1):119-23. doi: 10.1016/j.jbmt.2014.08.010. Epub 2014 Sep 3. PMID: 25603751.
 16. Michael G. Levitzky "Pulmonary Physiology" Seventh edition, chapter -1, page- 1 to 10
 17. Robert L. Wilkns, James k. Stoller, "Egan's Robert M. Kacmarek fundamentals of respiratory care" 9th Edition, Chapter-19, 399-430, chapter-23, 503-524
 18. K Sembulingam, Prema Sembulingam "Essentials of Medical Physiology" Fourth edition, chapter-118, page-609-612
 19. Chan-Yeung M, Ait-Khaled N, White N, Ip MS, Tan WC. *Journal Tuberc Lung Dis*. ""The burden and impact of COPD in Asia and Africa" (2004)
 20. W. Anees, P.F. Gannon et al "Effect of peak expiratory flow data quantity on diagnostic sensitivity and specificity in occupational asthma" *European Respiratory Journal* 2004
 21. Government Of India Ministry of Health And Family Welfare Department Of Health And Family Welfare Lok Sabha Unstarred Question No. 2243 To Be Answered On 23rd September, 2020
 22. Epidemiology: COPD in India (2011) Aging, population growth, and increased survival will drive a marked increase in prevalent cases of COPD over the next ten years. Available from www.datamonitor.com
 23. Wan c. Tan and Tze p. Ng (2008) "COPD in Asia Where East Meets West" (*CHEST* 2008; 133:517-527)

24. Nicholas R. Anthonisen et al (1994) "Effects of Smoking Intervention and the Use of an Inhaled Anticholinergic Bronchodilator on the Rate of Decline of FEV1." JAMA. 1994;272(19):1497-1505.doi:10.1001
25. Gavin C. Donaldson, et al (2005) "Airway and Systemic Inflammation and Decline in Lung Function in Patients With COPD." CHEST 2005; 128:199
www.chestjournal.org
26. NICE (National Institute for Health and Clinical Excellence) Chronic obstructive pulmonary disease: Evidence Update 5 - Chronic obstructive pulmonary disease (February 2012) Available from(www.evidence.nhs.uk/topic/chronic-obstructive-pulmonary-disease).
27. Garrod_ R, Lasserson . et al (2007) "Role of physiotherapy in the management of chronic lung diseases: an overview of systematic reviews." Respir Med. 2007 Dec;101(12):2429-36.
28. Banerjee G, Briggs M, Johnson MI. Kinesiology taping as an adjunct for pain management: a review of literature and evidence. Indian J Pain 2016;30:151–7
29. Petrovic P, Dietrich T, Fransson P, Andersson J, Carlsson K, Ingvar M. Placebo in emotional processing— induced expectations of anxiety relief activate a generalized modulatory Network. Neuron 2005;46:957–69.
30. Saniye & Daskapan, Arzu & Pekiavas, Nihan & Sakizli, Elif. (2018). Effects of Kinesio Taping Applied to Diaphragm Muscle on Aerobic Exercise Capacity and Pulmonary Function in Sedentary Individuals. Anadolu Kliniği Tıp Bilimleri Dergisi. 23. 10.21673/anadoluklin.385414.
31. Tomruk M, Keles, E, Özalevli S, Alpaydın AÖ. Effects of thoracic kinesio taping on pulmonary functions, respiratory muscle strength and functional capacity in COPD patients: a pilot randomized controlled study. EuroResp J 2017;50:A1534. DOI: 10.1183/1393003.congress-2017.PA1534
32. Langendoen, John & Fleish-man, Caren & Kim, Soon & An, Ho. (2016). Rationales and Evidence of Elastic Taping: A clinician’s per-spective. Journal of International Academy of Physical Therapy Research. 7. 979-988. 10.20540/JIAPTR.2016.7.1.979.
33. Ganesh, B. R.; Praneetha, N.1; Naik, Varun1. Comparison of Two Thoracic Kinesio-Taping Techniques on Pulmonary Function, Oxygen Saturation, and Exercise Capacity in COPD Patients: A Randomized Clinical Trial. Indian

Journal of Physical Therapy and Research 5(1): p 32-36, Jan–Jun 2023. | DOI: 10.4103/ijptr.ijptr_168_22

34. bartolome R. celli (2000) "The Importance of Spirometry in COPD and Asthma. Effect on Approach to Management" (CHEST 2000; 117:15S-19S)
35. T R Schermer, J E Jacobs, et al (2003) "Validity of spirometric testing in a general practice population of patients with chronic obstructive pulmonary disease (COPD)" Thorax 2003;58:861-866
36. Timothy j. Barreiro et al (2004) "An Approach to Interpreting Spirometry" AMERICAN FAMILY PHYSICIAN www.aafp.org/afp VOLUME 69, NUMBER 5 /MARCH 1, 2004
37. Stuart W. Stoloff et al (2010) "Diagnosis and treatment of patients with chronic obstructive pulmonary disease in the primary care setting: focus on the role of spirometry and bronchodilator reversibility" Supplement to The Journal of Family Practice | Vol 60,No 4| April 2011
38. Casanova C, Cote C, Marin JM, Pinto-Plata V, de Torres JP, Aguirre-Jaíme A, Vassaux C, Celli BR. Distance and oxygen desaturation during the 6-min walk test as predictors of long-term mortality in patients with COPD. Chest. 2008 Oct 1;134(4):746-52.
39. Ben Cheikh Rejbi I, Trabelsi Y, Chouchene A, Ben Turkia W, Ben Saad H, Zbidi A, Kerken A, Tabka Z. Changes in six-minute walking distance during pulmonary rehabilitation in patients with COPD and in healthy subjects. International journal of chronic obstructive pulmonary disease. 2010 Aug 9:209-15.
40. Australian Government Department of Health and Ageing, 15 July 2004.Chest Expansion Normal Data
41. Rufus Adesoji Adedoyin *Reference Values for Chest Expansion among Adult Residents in Ile-Ife" J Yoga Phys Ther 2012.
42. .Gan WQ, Man S, Senthilselvan A, Sin D. Association between chronic obstructive pulmonary disease and systemic inflammation: a systematic review and a metaanalysis. Thorax. 2004;59(7):574–580.
43. Sin DD, Man SF. Skeletal muscle weakness, reduced exercise tolerance, and COPD: is systemic inflammation the missing link? Thorax. 2006;61(1):1–3.

44. Donaldson AV, Maddocks M, Martolini D, Polkey MI, Man WD. Muscle function in COPD: a complex interplay. *Int J Chron Obstruct Pulmonary Dis.* 2012; 7:523–535.
45. O'Donnell DE, Webb KA. The major limitation to exercise performance in COPD is dynamic hyperinflation. *J Appl Physiol.* 2008;105(2):753–755.
46. De Troyer A. Effect of hyperinflation on the diaphragm. *Eur Respir J.* 1997;10(3):708–713
47. BAXI, Gaurang et al. Immediate Effects of Kinesio Taping on Lung Functions, Chest Expansion and Dyspnoea in COPD Patients. *Indian Journal of Physiotherapy & Occupational Therapy-An International Journal*, [S.l.], p. 101-105, mar. 2020. ISSN 0973-5674. Available at: <<https://www.ischolar.in/index.php/ijpot/article/view/193430>>
48. A study on kinesiology taping of diaphragm versus diaphragmatic breathing using progressive resisted exercise in asymptomatic young male smokers", *International Journal of Emerging Technologies and Innovative Research* (www.jetir.org | UGC and issn Approved), ISSN:2349-5162, Vol.5, Issue 6, page no. pp59-63, June-2018, Available at : <http://www.jetir.org/papers/JETIR1806252.pdf>
49. Daitx RB, Dos Santos K, Dohnert MB, da Silva TD, Silva JD. Limited utility of Kinesio Taping® in the physiotherapy treatment for patients with chronic obstructive pulmonary disease exacerbation. *Physiotherapy Theory Practice* 2018;34: 741–6.
50. Park SJ, Kim SH, Min KO. Effects of joint mobilization and respiratory muscle taping on pulmonary function and chest expansion ability in patients with chronic stroke. *J Int Acad Phys Therapy Res* 2017;8:1229–34
51. Malehorn K, Hiniker J, Mackey T, Heumann KJ, Murray SR, Pettitt RW. Kinesio Tape® applied to the thorax augments ventilatory efficiency during heavy exercise. *Int J Exercise Sci* 2013;6:157–63.
52. Ann Thomson, Alison skinner " Tidy's physiotherapy" 12th edition. Chapter 13.page-185-211.
53. Szczegieliński, Jan & Luniewski, Jacek & Bunio, Andrzej & Bogacz, Katarzyna & Śliwiński, Zbigniew. (2007). The use of kinesio taping in patients with acute stages of bronchial asthma. *Polish Journal of Sport Medicine.* 23. 337-341

54. Hombrados-Hernández, R. & Ortí, Eva & Buil, Miguel. (2011). Efectos de la aplicación del Kinesio taping™ en el diafragma en el resultado la cicloergoespiometría y la prueba marcha de 6 minutos. *Fisioterapia*. 33. 64-69. 10.1016/j.ft.2011.02.002.
55. Metin Ökmen, B., Şengören Dikiş, Ö., Ökmen, K., Altan, L., & Yildiz, T. (2019). Investigation of the effect of kinesiotaping on the respiratory function and depression in male patients with chronic obstructive pulmonary disease: a prospective, randomized, controlled, and single-blind study. *The Aging Male*, 23(5), 648–654. <https://doi.org/10.1080/13685538.2019.1567703>
56. HANY E. OBAYA, Ph.D., Lamis S. ABOALOYOUN, M.Sc.; and Nagy L. NASSEF, Ph.D.; MOUNIR H. BAHGAT, M.D.. “Effect of Kinesotaping on Vital Capacity in Patient with Chronic Obstructive Pulmonary Disease.” *The Medical Journal of Cairo University* (2022):
57. Padkao T, Boonla O. Relationships between respiratory muscle strength, chest wall expansion, and functional capacity in healthy nonsmokers. *J Exerc Rehabil*. 2020 Apr 28;16(2):189-196. doi: 10.12965/jer.2040080.040. PMID: 32509705; PMCID: PMC7248446.
58. Shah M, Julu POO, Monro JA, Coutinho J, Ijeh C, Puri BK. Neuromuscular taping reduces blood pressure in systemic arterial hypertension. *Med Hypotheses*. 2018 Jul;116:30-32. doi: 10.1016/j.mehy.2018.04.014. Epub 2018 Apr 17. PMID: 29857904.

ANNEXURE

ANNEXURE-1

CONSENT FORM

STUDY TITLE; “IMMEDIATE EFFECT OF CHEST WALL KINESIO TAPING ON PULMONARY FUNCTIONS, CHEST EXPANSION AND SIX MINUTE WALK DISTANCE IN INDIVIDUALS WITH CHRONIC OBSTRUCTIVE PULMONARY DISEASE.”- EXPERIMENTAL STUDY

Name of Participant:

સહભાગી નું નામ:

Address:

સરનામું:

Contact no :

ફોન નંબર :

I, _____ give consent for my participation in this study. All the aspects of study dangers have been explained & all my questions have been answered to my satisfaction in my understandable language. I understand that there is no predictable risk in participating in the study. I understand that I can withdraw from study at any time without giving reason and without any penalty & that will not affect my future visits to this department.

હું, _____ સદર અભ્યાસ માં ભાગ લેવા સંમતિ આપું છું. આ અભ્યાસ ના સંભવિત જોખમી પાસાઓ વિષે મને જાણ કરવામાં આવી છે, અને મારા દરેક પ્રશ્નોના જવાબો મારા સંતોષ મુજબ હું જે ભાષા જાણું/સમજું છું તેમાં અપાયા છે. હું સમજું છું કે સદર અભ્યાસમાં ભાગ લેવામાં મને કોઈ પણ જાતનું સંભવિત જોખમ નથી. હું સમજું છું કે આ અભ્યાસમાંથી કોઈપણ સમયે કારણ દર્શાવ્યા વિના કે કોઈ પણ દંડ વગર હું મારી જાતને પાછી ખેંચી શકું છું અને તે મારી ભવિષ્યની આ વિભાગની મુલાકાતને અસર કરશે નહીં.

SIGNATURE OF THE PARTICIPANT `

ભાગ લેનારની સહ

ANNEXURE-2

PROFORMA

Sr.no:

PERSONAL DETAILS:

NAME:

AGE:

GENDER:

HEIGHT:

WEIGHT:

BMI:

OCCUPATION:

REF BY:

CHIEF COMPLAINT:

LIFESTYLE:

SMOKER/ EX-SMOKER/ NON-SMOKER / OTHERS

HISTORY:

PRESENT HISTORY:

PAST MEDICAL HISTORY:

DRUG HISTORY:

SURGICAL HISTORY

FAMILY HISTORY:

PERSONAL HISTORY:

SUBJECTIVE ASSESSMENT:

BREATHLESSNESS:

COUGH:

WHEEZE:

SPUTUM / HEMOPTYSIS:

CHEST PAIN:

OBJECTIVE ASSESSMENT:

ON OBSERVATION:

ON EXAMINATION:

ON PERCUSSION:

ON AUSCULTATION:

SKIN EXAMINATION: - 2minute patch test (over forearm)

COPD GOLD CLASSIFICATION GRADE I & II

INVESTIGATION: PRE-VITALS POST VITALS

HEART RATE		
BLOOD PRESSURE		
RESPIRATORY RATE		
SPO2		
MIDDLE CHEST EXPANSION		
6 MWD		

PULMONARY FUNCTION TEST

COMPONENTS	PRE	POST
FVC		
FEV ¹		
FEV ¹ / FEC		
PEFR		

MASTER CHART

Sr. No	AGE	PRE FEV ₁	POST FEV ₁	PRE FVC	POST FVC	PRE FEV ₁ /	POST FEV ₁ /	PRE PEFR	POST PEF	PRE 6MW	POST 6MW	PRE MCE	POST MCE	PRE SPO ₂	POST SPO ₂	PRE RR	POST RR	PRE HR	POST HR	PRE MBP	POST MBP
1	52	59	65	63	65	78	80	98	96	390	405	2.5	3.2	95	96	18	19	79	80	101	105
2	56	60	65	67	72	60	65	100	89	300	330	0.5	1	97	97	18	19	85	82	83	91
3	58	68	75	63	64	62	68	69	79	345	360	1	1	96	97	19	20	76	77	98	100
4	49	60	68	68	70	63	69	49	48	330	345	1.8	2.3	95	95	20	20	79	78	112	125
5	55	64	69	55	57	68	71	55	56	360	375	2	2.4	96	97	19	20	69	70	106	103
6	48	64	70	68	69	65	67	48	51	390	405	2.5	2.52	97	99	19	18	79	80	103	107
7	51	67	72	55	57	54	55	49	52	420	435	2	2.5	95	96	18	17	65	70	100	105
8	58	69	77	48	52	56	55	100	105	300	330	1.5	2	98	98	18	17	80	82	105	105
9	41	62	69	67	66	72	75	73	77	315	330	2	2.1	96	98	18	17	82	80	93	107
10	54	66	71	50	54	62	69	60	68	345	360	1.8	2	97	98	19	18	77	78	127	113
11	56	70	77	74	69	55	59	46	48	375	390	3.2	4.8	96	97	20	19	75	74	83	95
12	59	63	70	61	79	45	51	81	80	405	390	0.8	1.5	96	97	21	20	75	74	80	95
13	60	69	77	61	79	49	58	61	65	345	375	1.5	1.5	95	96	18	19	82	80	105	115
14	58	65	70	78	79	61	70	84	76	315	345	1.5	2.3	96	97	19	18	68	70	97	105
15	55	55	63	65	66	60	62	72	69	375	405	2	2.2	97	96	20	19	80	82	103	100
16	49	52	58	69	72	67	63	82	79	345	360	2.3	2.5	95	96	21	20	82	82	97	95
17	60	73	78	72	71	76	76	59	55	420	435	2	2	98	99	18	17	80	78	83	97
18	48	65	72	76	77	69	72	60	71	390	420	2.5	2.5	95	96	19	18	65	67	100	105
19	58	63	68	62	70	53	58	68	72	435	450	2	2.5	96	98	20	19	70	73	93	105
20	52	59	66	70	72	48	52	80	95	330	360	1.5	2	98	99	21	20	80	83	97	100
21	50	67	72	53	55	50	61	90	96	345	360	3	3.2	95	96	18	17	79	82	101	98
22	53	56	63	61	63	53	59	83	96	390	405	2.5	3	95	97	19	18	85	85	83	94
23	60	55	62	49	52	70	72	100	105	330	330	0.8	2	94	96	18	18	80	85	77	90
24	49	65	71	58	62	59	65	112	116	420	435	1	1.8	94	97	19	18	72	75	97	99
25	58	71	75	61	64	42	50	108	119	345	375	2	2.3	95	98	20	19	69	72	103	105
26	60	58	65	59	63	49	55	90	99	405	405	2.6	3.2	96	97	21	20	74	76	90	92
27	56	54	62	64	68	50	68	89	98	300	330	1	1	97	98	19	18	76	79	83	87

